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Master Thesis

Pronominal Gender Bias in Large Language Models with Language Adapters

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Abstract

Language adapters have gained recognition for high portability and parameter efficiency for cross-lingual transfer learning (Pfeiffer, Vulić, et al., 2020). However, their gender bias characteristics are still under-explored. We, therefore, evaluate gender bias in English (Rudinger et al., 2018) and Chinese using parallel Winogender datasets, the last of which we have constructed ourselves. By utilizing this multilingual diagnostic corpus to detect gender bias, we first directly evaluate and compare the monolingual language models and their respective multilingual versions for the tasks of pronoun prediction and coreference resolution. Next, our primary focus shifts to multilingual models, where we train language adapters and conduct full fine-tuning on Wikipedia data for language modeling. We then compare the results of the *adapter* and *finetune* model versions for pronoun prediction and coreference resolution with that of the *base* version that has not been trained additionally for any task, and also identify and compare potential gender bias across the three model versions. From our evaluation and comparison, we summarize that in our specific settings, 1) multilingual models perform worse than monolingual models in general; 2) additional training for language modeling usually introduces more gender bias but does not necessarily improve model performance. However, to reliably quantify and evaluate model performance and gender bias, we still need to consider multiple factors about training data and task, prompt quality, and automated evaluation of gender bias and performance.

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Chapter 1.

Introduction

Currently, the primary approach to carrying out downstream Natural Language Processing (NLP) tasks involves fine-tuning large pre-trained language models (Devlin et al., 2019; Radford et al., 2019). Training a separate LM for each specific downstream task and storing a complete set of tuned LM parameters is both computationally expensive and space-inefficient. For instance, mT5 comprises from 300 million to 13 billion parameters (Xue et al., 2021). To alleviate such a problem, Houlsby et al. (2019) proposed *Adapters*, a compact and extensible framework that inserts additional task-specific layers into the layers of pre-trained LMs. This approach attains comparable performance as fine-tuning the model on language understanding and generation tasks while only adding around 3% parameters (Houlsby et al., 2019; Lin et al., 2020). Pfeiffer, Vulić, et al. (2020) developed an adapter-based framework, MAD-X, enabling parameter-efficient cross-lingual transfer.

Fine-tuning and adapters are very popular in multilingual settings to make models specialize in certain languages. Nevertheless, to the best of our knowledge, most prior studies revolve around the aspect of performance and other aspects such as gender bias characteristics are still under-explored and have not been really quantified. Inspired by this, this research will concentrate on the impact of adapter modules on pronominal gender bias in language models compared with full fine-tuning. More precisely, we will shed light on the following research questions:

- What is the impact of fine-tuning and adapters on pronominal gender bias?
- What pronominal gender bias characteristics are maintained, minimized, or amplified by the use of language adapters? And what is the extent?
- Will English and Chinese adapters result in similar or different influences on the bias?

In order to answer these questions, we first build a Chinese Winogender dataset parallel to English Winogender Schemas (Rudinger et al., 2018). Then, we train an English and a Chinese language adapter for masked language modeling using Wikipedia data, and also fully fine-tune multilingual Masked Language Modeling (MLM) models with the same data. Based on this, we conduct experiments for the two downstream tasks of pronoun prediction and coreference resolution. We adopt masked language modeling in both of the tasks and also prompt engineering in coreference resolution. Our approach initially involves an evaluation and comparison of pre-trained monolingual language models and their respective multilingual counterparts. After that, we shift our focus only to the multilingual models and conduct a comparative analysis of the results and bias characteristics exhibited across three versions: *base* version (without any training), *adapter* version (combined with a trained language adapter), and *finetune* version (fully fine-tuned).

Our contributions are mainly two-fold:

- We have established a Chinese Winogender dataset that closely parallels the English Winogender Schemas. We also extend the original dataset to various variants such as single-entity and no-pronoun versions, which can achieve different experimental purposes. The creation of this parallel Chinese dataset enables a comprehensive exploration of gender bias, helping us understand bias across different languages.

- Our study goes beyond evaluating models with language adapters compared with fully fine-tuned models only in terms of performance. Besides that, we also explore pronominal gender bias reflected in models with adapters and compare it with that in pre-trained and post-finetuned models.

Our findings are: 1) based on the results of pronoun prediction, the multilingual MLMs seem to contain more internal pronominal gender bias than the monolingual English MLMs but similar to the monolingual Chinese MLMs, and these MLMs are inherently male-oriented. 2) monolingual models outperform multilingual ones overall in coreference resolution, but both of them are very bad at this task in zero-shot transfer, and prompt quality has a big impact on the results; 3) based on the performance gaps among genders in coreference resolution, compared with the *base* version, the English adapter could minimize pronominal gender bias but inhibits model performance and the Chinse adapter would introduce more bias from training data and also hurt model performance; 4) compared with the *finetune* version, both adapters show worse performance but lower bias.

Chapter 2.

Related Work

2.1. Adapter

Adapter modules are applied as lightweight tuning where most pre-trained parameters are fixed and the pre-trained model is modified only with small trainable modules in downstream tasks. Adapter tuning is a parameter-efficient approach that changes the model behavior by modifying the activations at any given layer instead of adding new input representations. Houlsby et al. (2019) constructed the bottleneck adapters and brought this framework to the NLP field with performance close to fully fine-tuned models in natural language understanding tasks. To mitigate catastrophic forgetting, *AdapterFusion* was proposed by Pfeiffer, Kamath, et al. (2021). This algorithm is composed of knowledge extraction and knowledge composition and it can leverage knowledge better from multiple tasks. With its development to cross-lingual learning, Rezaee et al. (2021) deployed adapters to transfer a pre-trained monolingual model to an unseen language by learning new token-level embeddings. Pfeiffer, Vulić, et al. (2020) used multiple adapters in a multilingual context to explicitly separate language understanding from task specification and reached competitive results in zero-shot cross-lingual transfer. Shortly afterward, Pfeiffer, Vulić, et al. (2021) further explored novel data-efficient methods that resulted in substantial performance gains for resource-poor languages with unseen scripts in model pre-training. Our research concentrates on language adapters with the same configuration as MAD-X (Pfeiffer, Vulić, et al., 2020).

2.2. Gender bias

Gender bias in language models has been studied a lot in extensive prior work. Language models are very likely to reflect and amplify existing cultural prejudices and inequalities by replicating human behavior and perpetuating bias (Sweeney, 2013). Gender bias has been shown to exist in word embeddings

(Bolukbasi et al., 2016; Garg et al., 2018; Swinger et al., 2019; Zhao, T. Wang, Yatskar, Cotterell, et al., 2019), as well as in a wide spectrum of models designed for downstream tasks such as machine translation (Savoldi et al., 2021) and sentiment analysis (Kiritchenko and Mohammad, 2018). Coreference resolution is a typical downstream task for gender bias due to various aspects of gender embedded in coreference inferences. For example, the gender property is shown explicitly and stereotypes or expectations around gender roles would be assumed (Cao and Daumé III, 2021). Zhao, T. Wang, Yatskar, Ordonez, et al. (2018) and Rudinger et al. (2018) designed the WinoBias and the Winogender schemas respectively with sample sentences in the style of the Winograd Schema. These two diagnostic datasets are widely used to detect and measure gender bias in language models.

Most majority the previous research only revolved around binary gender and did not deal with the broader field of identity inclusion. Dev et al. (2021) investigated harms associated with the treatment of gender as binary and how these harms are captured and perpetuated in language representations. Recent studies (Cao and Daumé III, 2021; Lauscher et al., 2022) have expanded the definition of gender and shed light on the challenges of non-binary gender identities in NLP systems. Monolingual focus is another limitation in this field. Current research on gender bias is usually monolingual, focusing predominantly on English or a few high-resource languages like Spanish (Zmigrod et al., 2019). It is necessary to map such multilingual margins as gender bias is derived from our cultural and societal perspectives and may vary across languages (Stanczak and Augenstein, 2021). We will examine pronominal gender bias in language adapters with a multilingual setup.

We need gender bias metrics to measure gender bias and they are usually divided into two groups, extrinsic and intrinsic metrics. Extrinsic metrics measure gender bias concerning a specific downstream task and intrinsic metrics measure bias in internal model representations and are not directly related to any downstream task. On the intrinsic side, foundational metrics include the Word Embedding Association Test (WEAT) developed by Caliskan et al., 2017, to quantify bias in static word embeddings. The Sentence Encoder Association Test (SEAT) proposed by May et al., 2019, extends this assessment to sentence embeddings to capture contextualized bias. Moreover, the Contextualized Embedding Association Test (CEAT) by Guo and Caliskan, 2021 provides another contextual alternative to WEAT, bringing us a nuanced exploration of bias in language models.

On the extrinsic side, Orgad and Belinkov, 2022 summarized the current paradigm of gender bias evaluation. Extrinsic metrics can be further classified into two subcategories: extrinsic performance metrics and extrinsic predic-

tion metrics. Extrinsic performance metrics assess how a model's performance is influenced by gender, usually calculated with specific gold labels. On the other hand, extrinsic prediction metrics evaluate a model's predictions such as output probabilities. They are irrelevant to gold labels and bias is measured by analyzing the impact of gender or stereotypes on the model's predictions. However, these extrinsic gender bias metrics are limited to binary genders. Our approach extends beyond these metrics and introduces gender-inclusive measures for binary and neutral gender.

Chapter 3.

Experimental Methodology

3.1. Task

We perform two downstream tasks to detect pronominal gender bias, one is pronoun prediction and another is coreference resolution.

3.1.1. pronoun prediction

In the pronoun prediction task, we will adopt MLM models and mask the pronoun in the input sequence directly, and then make the models to predict this masked token. For this task, our data is the Winogender Schema (Rudinger et al., 2018) and its variants, which will be described in detail in the next section. Due to the specificity of the data, there is no standard correct answer for this task. Here is a simple example, the output for the masked token could be pronouns *she/he/they* or any other single word. Hence, we do not use any exact metrics for this task and only obtain descriptive results which will be elaborated on in Chapter 5.

- The technician told the customer that **[MASK]** could pay with cash.

3.1.2. coreference resolution

Coreference resolution systems usually solve the task by detecting mentions through text spans and then predicting for each pair of spans if they represent the same entity, but we apply prompt engineering to this task in our research. The data are prompted using various types of templates and then presented to the model letting it resolve which entity the pronoun refers to. Unlike the pronoun prediction task, we do have correct answers here. No matter the gender of the pronoun, it always refers to one entity according to the meaning of the sentence, and this is why we can use the actual metrics in this task. Like in Example 2, the pronoun can only refer to "the customer" or "someone" based on the meaning of this sentence, not "the technician". The detailed evaluations

will be explained in Section 3.5 and the qualitative results will be reported in Chapter 6.

3.2. Data

In our research, we mainly adopt a multilingual diagnostic corpus in English and Chinese to detect gender bias. Based on the English Winogender benchmark (Rudinger et al., 2018), we extend the original data to different version variants and also build the Chinese parallel dataset on our own.

3.2.1. English Winogender Schema

The English Winogender Schema is a diagnostic dataset to identify gender bias in coreference resolution systems. It is designed not to pose a difficult challenge for the task of coreference resolution, but to detect possibly existing gender bias in NLP systems and only serve as a test set. But it can also be used to assess the capabilities of Natural Language Understanding as one of the components of SuperGlue (A. Wang et al., 2019). The original English Winogender schemas contain 120 templates. We display the following instances to understand this dataset clearly. From Example 1, we can see that each template includes three key components. According to the explanation in the paper of Rudinger et al. (2018), $OCCUPATION is a person referred by their profession; $PARTICIPANT is a secondary person or participant; $PRONOUN is coreferent with either by $OCCUPATION or $PARTICIPANT. There are three types of pronouns depending on grammatical requirements: nominative, possessive, and accusative. Note that all genders (female/male/neutral) make sense and the gender property does not influence the resolution because all templates are unambiguous for humans based on their semantics. Therefore, as Example 2 shows, each template gives rise to two triple items by replacing a specific participant with the generic participant "someone". The only difference among the triple sentences is the pronoun gender. As a result, there are 720 (120x2x3) sentences in total.

1. The **$OCCUPATION** told the **$PARTICIPANT** that **$PRONOUN** could pay with cash.
2. The **technician** told the **customer/someone** that **she/he/they** could pay with cash.

We also make some modifications to the original Winogender Schema to avoid ungrammatical sentences when the neutral pronoun is applied. For in-

stance, we change "$PRONOUN was repairing the sidewalk" to "$PRONOUN would repair the sidewalk". This ensures that the sentence is grammatically correct regardless of the gender of the pronoun.

3.2.2. Different version variants

Based on the original data, we have derived the **no-pronoun** and the **single-entity** versions to meet different experimental needs.

No-pronoun As the name suggests, each sentence in this version excludes the pronoun and we replace it with the entity itself that it refers to. The purpose of generating this version is to see if the model is able to predict correctly when the referent entity is already known. The prerequisite for detecting gender bias based on the original data is that the model is capable of coreference resolution. To verify this, we need to make the above changes like from Example 2 to Example 3 to the original data and get a no-pronoun version.

3 The technician told the customer/someone that the **customer/someone** could pay with cash.

Single-entity Unlike the original sentence which contains two entities, this version contains only the pronoun and the single entity to which the pronoun refers. We compare the results obtained from this version in the coreference resolution task as a baseline with the results of the original version. This can help us quantify the extent to which the model is confused by the double entities.

4 **The technician told the customer that she/he/they** could pay with cash.
The technician told the customer that she/he/they had completed the repair.

5 **The customer was told** that she/he/they could pay with cash.
The technician said that she/he/they had completed the repair.

However, the Winogender Schema was originally constructed so that both versions of the sentence are exactly parallel with each other until the pronoun, but our single-entity cannot maintain this anymore. Example 4 is one parallel sentence pair from the original Winogender Schema and Example 5 is one sentence pair from the single-entity version. We can see that the first part in Example 4 remains the same, but the meaning completely changes in Example 5 and not all entities appear in the same context in this single-entity

version. Accounting for this, we propose the following three types of single-entity alternative templates to equalize conditions for all occupations and participants across the major pronoun types (nominative, accusative, possessive). Therefore, these alternatives are a truly parallel way of comparing different pronouns, occupations and participants.

- The $OCCUPATION / $PARTICIPANT said that $NOM_PRONOUN felt happy.
- The $OCCUPATION / $PARTICIPANT said that $POSS_PRONOUN day went well.
- The $OCCUPATION / $PARTICIPANT said that sunny days bring $ACC_PRONOUN joy.

To summarize the usage of our different data variants, we will only use the single-entity version and the alternatives in the pronoun prediction task. This is because we simply want to detect and observe the gender bias from predicted probabilities of pronouns of the three genders, so we try to remove the influence from coreference resolution with double entities. The original data (or double-entity), the single-entity and the no-pronoun versions will be used in the coreference resolution task. We also construct the parallel Chinese Winogender dataset on the basis of all English versions. More details will be introduced and explained in Section 4. The usage of Chinese data is the same as English.

3.3. Model

To perform the tasks of pronoun prediction and coreference resolution, we employ two types of pre-trained large language models, Masked Language Modeling (MLM) models and generative models. The former is pre-trained on large text corpora and is designed to predict masked or hidden words within a sentence, while the latter aims to generate coherent and contextually relevant text. The selected language models in our research are consistent with the base size. In instances that are case-sensitive, we uniformly use the uncased versions.

3.3.1. Masked Language Modeling models

We choose to utilize well-known monolingual and multilingual MLM models. We experiment with the English data using Bidirectional Encoder Representation Transformers (BERT) (Devlin et al., 2019) and robustly optimized BERT

(RoBERTa) (Liu et al., 2019). As for the Chinese data, Cui et al. (2020) revisited and released Chinese pre-trained language models which have been shown to achieve great performances on many NLP tasks, so we select Chinese BERT and Chinese RoBERTa with Whole Word Masking (BERT-wwm and RoBERTa-wwm). This means that the whole Chinese word instead of individual Chinese characters is masked in the pre-training phase, so that the WordPiece tokenizer no longer splits the word into small fragments as before because Chinese words are not formed by alphabet-like symbols but are composed of characters (Cui et al., 2020). We also use their respective multilingual versions, mBERT (Pires et al., 2019) and XLM-RoBERTa (Conneau et al., 2020), to further compare the performance of the monolingual and multilingual models.

Besides these untrained base models, we will fine-tune the multilingual ones and train language adapters, and then combine the adapters with the base models to perform both downstream tasks as introduced in Section 3.1. We use the script for masked language modeling[1] from the *Adapters* library to train language adapters and fine-tune models. The hyperparameter setting will be displayed in Appendix C.

3.3.2. Generative Models

For generative models, we use Text-To-Text Transfer Transformer (T5) (Raffel et al., 2020) and its multilingual version mT5 (Xue et al., 2021) in base size. Originally introduced by Raffel et al. (2020), the T5 model is pre-trained on a number of NLP tasks via a unified framework using a text-to-text format. In contrast, the multi-lingual T5 model is pre-trained on 101 languages from a CommonCrawl-based corpus, including Chinese and English. FLAN, an enhanced version of T5 that has been finetuned in a mixture of tasks by Chung et al. (2022), is also in our generative model list. FLAN-T5 is better than T5 with the same number of parameters after being fine-tuned on over 1,000 more tasks such as definite pronoun resolution. Therefore, we expect it to excel in downstream tasks. As for Chinese generative models, there is no specific Chinese T5, but we find a small Chinese chat model developed by Chen (2023) based on the T5 with only 0.2B parameters, ChatLM. We will experiment with these generative models only for coreference resolution.

[1] https://github.com/adapter-hub/adapters/blob/main/examples/pytorch/language-modeling/run_mlm.py

3.4. Methods

3.4.1. Masked Language Modeling

We employ Masked Language Modeling to do pronoun prediction and coreference resolution with the help of prompting. It begins with the tokenization of the input text. After getting tokenized representations, some of them are randomly selected and replaced with a special masked token (the masked token should be pre-defined in the input text if aiming at a specific word). A label tensor used for loss calculation and optimization is then generated from the input representation. During training, the model is trained to minimize a loss function that measures the difference between its predicted tokens and the actual masked tokens. The key innovation of models like BERT is their ability to consider the bidirectional context, both the words to the left and right of the masked token, when predicting a masked token.

It should be noted again that although BERT-wwm and RoBERTa-wwm have used Whole Word Masking, this only affects the selection of the masking tokens in the pre-training. The WordPiece Tokenizer will still split a Chinese word into several different parts when performing downstream tasks. As explained in Section 3.3, this is because a Chinese word usually includes more than one character. Therefore, in order to obtain more reliable and accurate predictions, when masked language modeling Chinese data, the number of masks we insert is consistent with the number of characters in the sentence that need to be masked. In other words, the input sequence may contain multiple mask tokens.

3.4.2. Prompting

Prompt engineering is a technique used to guide the behavior of language models by providing them with specific instructions or queries in the form of a prompt. Obviously, crafting carefully designed prompts is essential in this method, because prompts can instruct the model on the desired task or the type of response it should generate and directly influence output quality and the model's performance. These prompts can take various forms, such as questions, fill-in-the-blanks, or directives. They may include explicit and clear instructions, like specifying the format of the answer, asking the model to think step by step, or providing context to narrow down the response. Developers usually experiment with various prompts iteratively to optimize the performance of the intended task. Prompting allows us to leverage pre-trained models for specific applications by tailoring their behavior through well-crafted prompts and we will make full use of it for coreference resolution.

Referred to the study of Sanh et al., 2021, the initial prompt templates we select revolve around yes/no and options (which are called *Yes or No* and *Option* respectively since now). Unfortunately, the overall performance is worse than we expected and there are some issues with these two types of templates based on our preliminary experiments. For example, pre-trained language models love to predict "no" for literally everything in English data and prefer to predict "是 (yes)" in Chinese data. To solve those problems, we generate the templates that instruct the model to predict directly (which are called *Direct*) based on *Yes or No*. Inspired by Kotek et al., 2023, we also add another *Who* template including a follow-up "who" question about the action in the original sentence. To make sure the grammatical sentences of our data, we also review all of the "who" questions manually. All of these prompt templates are displayed in detail in Appendix A. When using the single-entity version, *Option* will not be used since there is only one entity in the sentence but two options are needed. For the no-pronoun data version, we only use *Who* for comparison with the normal double-entity version.

3.5. Evaluation

We will perform two downstream tasks as introduced in Section 3.1. However, in the prediction pronoun task, there is no unique measure for the prediction results, so the predictions will not be analyzed with accurate numbers but with illustrations about gendered occupations in Chapter 5. For the coreference resolution task, both MLM models and generative models are tested and we do have a standard answer. This section will explain how we evaluate model performance in this task.

3.5.1. Metrics

We experiment with masked language modeling and generative models in coreference resolution. Generally speaking, for the MLM models, we examine whether word prediction exactly matches the label or not to obtain accuracy and F1 scores, while for generative models, the accuracy is computed by the standard if generation includes either the occupation or the participant but not both of them.

The outputs from the MLM models are single tokens based on the masked tokens in the input sequences. For Chinese data, unlike the inserted single mask token in one English input sequence, one Chinese input sequence can contain more than one mask token, thus resulting in more than one corresponding prediction. Since each mask token produces a single Chinese character, we

choose to concatenate all the predicted characters together as the final result for comparison with the label during evaluation. After obtaining the raw predictions, we still pre-process them by lowering all the letters. Next, we will calculate the accuracy and F1 score for each prompt template for further analysis later. These two metrics are computed based on whether the characters in the model's prediction exactly match those in the gold label; otherwise, it will be judged as a wrong case. But there is still an exception. We observe that in the predictions of English data, the outputs using *Option* prompts are heavily populated with the word "former" or "later". For such meaningful and reasonable outputs, we relax the evaluation criteria for English test data by replacing "former" or "later" with the corresponding $OCCUPATION or $PARTICIPANT in the original input sentence, and then compare this meaningful referred entity with the label for subsequent computations.

For the generative models, the outputs are continuous text. It is inappropriate to still use the criterion of whether the prediction exactly matches the label or not. Instead, our standard has been changed to examine whether the label exists in the generated text, and $OCCUPATION and $PARTICIPANT must not be present at the same time. If the generation meets these two requirements, then it is considered to be correct. We calculate the accuracy and F1 score based on this.

Since the generations contain a lot of noise, such as some meaningless punctuation marks or even the mask token of the T5 model "<extra_id_0>", we adopt different pre-processing steps according to different language characteristics. For English generations, we first convert all text to lowercase, then remove all punctuation marks, and split consecutive words into separate tokens to avoid misjudgments. For example, when evaluating *Yes No* prompts, the label "no" is a very common letter combination and can be included in many words like "pronoun". If we do not split the generation into a list of single tokens, then it will be judged wrongly because "no" is indeed included in the string "pronoun". Therefore, the splitting step is necessary for English generations. For pre-processing Chinese generations, there is no case difference in Chinese, but punctuation still exists. Since the data used to pre-train the models contained a small portion of English data, even English words can appear in some outputs. So we only extract the Chinese text from the raw generations for evaluation.

In short, we have tried our best to develop evaluation methods in conjunction with the model outputs as much as possible. However, this is an automated evaluation and no manual judgment can be applied to make an accurate assessment, which leads to some special cases being ignored. We will discuss this further in Chapter 6 and Chapter 7.

3.5.2. Verbalized & Non-verbalized

For masked language modeling models, we obtain two types of outputs. One is non-verbalized, which is the raw output with the highest likelihood from the model as usual. Another is verbalized, which is selected from the limited options. To be more specific, the verbalized prediction is either participant or occupation with a higher probability, so we need to extract the probabilities of the options and compare them first.

The motivation for getting two different types of predictions is that there are some special cases but our evaluation methods cannot cover all of them as explained above, so we decide to constrain the range of outputs to be one of the entities in the input sentences. We expect that the evaluated performance based on the verbalized predictions would be better than on the non-verbalized predictions. The limited options include true labels and there are only two choices and this helps the models get a 50% chance of being correct. The more detailed comparisons will be reported and analyzed in Chapter 6.

3.6. Gender Bias Evaluation

To measure gender bias, we adopt extrinsic metrics in our research for pronoun prediction and coreference resolution. As explained above, only descriptive results are available for the task of pronoun prediction because no gold labels exist in this task. Therefore, we cannot quantitatively measure gender bias based on performance, but plot probability distributions using stacked bar graphs. These graphs are directly showing gender bias. Still, in order to confirm our abstract perceptions obtained from observing the graphs, we propose a quantitative calculation that aids in measuring bias to some extent. More details will be elaborated on in Chapter 5.

As for coreference resolution, we can evaluate model performance using specific metrics (accuracy and F1 scores) because there are fixed correct answers for testing sentences. The performance gap serves as a measure of how a model's performance is influenced by gender. In our Winogender corpus, where the only difference in each set of minimal triple sentences is the pronoun gender, this metric is very suitable. Researchers usually measure the difference in some performance metric for binary genders between male and female examples across the entire dataset. However, our coreference resolution task is gender-inclusive, introducing an additional non-binary gender, neutral. Accounting for this aspect, we extend the gender bias metric to performance gap for the three genders.

Specifically speaking, we first calculate an overall unweighted average performance across all gender groups, as the number of samples for each gender in the testing set is identical and the data is well-balanced. Then we assess the performance gap for each gender group by comparing its performance metric to the average performance. Lastly, the performance gaps are aggregated through averaging to get a comprehensive measure of bias covering all genders. A higher gap represents more gender bias. This approach is a nuanced and inclusive evaluation of gender bias in the model's performance for coreference resolution with non-binary gender.

Chapter 4.

Chinese Winogender Dataset

Completing a parallel Chinese version of Winogender schemas is one of the biggest contributions of this project. This chapter will revolve around how we construct the Chinese Winogender dataset and how we extend it to different versions as mentioned in Section 3.2.

In building the Chinese Winogender dataset, we mainly follow the following three principles (in descending order of importance):

1. Regardless of the gender of the pronouns, there is one and only one pronoun resolution. The meaning of each sentence template must be clear, and the words used must be unambiguous (which is also the aim pursued in drafting the English Winogender Schemas (Rudinger et al., 2018));
2. Keep parallelism with the original English sentence templates as far as possible, but it is still allowed to change the types of pronouns according to the context if necessary. For example, we can make some adjustments when the sentence is ambiguous or does not conform to Chinese grammar;
3. While ensuring the first two principles, localize the occupations and related example sentences to make them more in line with the Chinese context, more natural in tone, and more authentic in words.

Based on these three principles, we generated two versions of Chinese data based on the original English data: double-entity and single-entity, which will be described in detail next.

4.1. Double-Entity

As the name suggests, all sentence templates include two entities (occupation and participant) in this version. Our first step is to translate the English templates to Chinese ones using Google Translate API [1], then manually check and

[1] https://cloud.google.com/translate/docs/reference/api-overview

correct errors in the translated templates because some Chinese translations in specific contexts are inappropriate or even ungrammatical. Some errors are caused by different linguistic characteristics such as no definite clause in Chinese, thus removing the repeated participant token which occurs twice in one template. After that, we use the same approach to extend the Chinese templates to six test items: two ways of participants (a descriptive noun and 某人 (someone)) × pronouns with three genders. Finally, we check and modify them carefully again.

There are also three corresponding pronouns in Chinese, nominative pronouns, accusative pronouns and possessive pronouns, as follows:

- Nominative pronouns: 她 (she), 他 (he), *TA* (they)
- Accusative pronouns: 她 (her), 他 (him), *TA* (them)
- Possessive pronouns: 她的 (her), 他的 (his), *TA* 的 (their)

It is worth noting that there is no official non-binary pronoun in Chinese and we decide to use the relatively most common one "TA". However, it is still rare for language models, thus influencing the results of neutral pronouns negatively or even may not work with all modeling scenarios. As shown above, since all Chinese pronouns have only one pronunciation "tā", there are no grammatical problems with neutral gender pronouns, such as singular or plural nouns and verbs. Since all Chinese pronouns have only one pronunciation, there will be no grammatical problems with neuter pronouns in English, such as singular and plural conjugations of verbs and nouns.

For referred entities with close meanings, we deliberately distinguish them during translation to ensure that the entities in Chinese and English uniquely map to each other. For example, the occupations "practitioner" and "physician" are not clearly differentiated in Chinese translation, but we have adopted a more nuanced translation that can discern the tiny differences. Appendix B displays a dictionary of all of the Chinese and English entities in the Winogender Schema.

Last but not least, to guarantee the quality of the data and the unique clarity of the answers for reference resolution, three native Chinese speakers help us review the sentence samples one by one. In the end, all sentences pass the quality check. The following two examples are English and Chinese parallel sentences in the double-entity version.

The **technician**	told	the **customer**	that **she/he/they** could pay with cash.
技术员	告诉	**顾客**	**她/他/TA** 可以用现金支付。

4.2. Single-Entity

Following the three important principles mentioned above, the generation process of this version is basically the same as the double-entity version, but it is based on the English single-entity version that we have intentionally modified rather than the original English Winogender sentences. Here are the single-version parallel sentences based on the examples in Section 4.1.

The **customer**	was told	that **she/he/they** could pay with cash.
顾客	被告知	**她/他/TA** 可以用现金支付。

Besides, we also generate three types of Chinese single-version alternatives parallel to the English ones. These sentence templates are also not constrained by professional status in the Chinese context. They are:

1. $OCCUPATION / $PARTICIPANT 说 $NOM_PRONOUN 感到开心。($OCCUPATION/$PARTICIPANT said that $NOM_PRONOUN felt happy.)
2. $OCCUPATION / $PARTICIPANT 说 $POSS_PRONOUN 一天过得很好。(The $OCCUPATION/$PARTICIPANT said that $POSS_PRONOUN day went well.)
3. $OCCUPATION / $PARTICIPANT 说晴天带给 $ACC_PRONOUN 喜悦。(The $OCCUPATION/$PARTICIPANT said that sunny days bring $ACC_PRONOUN joy.)

The original single-entity version will be used in both pronoun prediction and coreference resolution, and the alternatives will only be experimented with in pronoun prediction.

Chapter 5.

Pronoun Prediction

To get descriptive results for this task, we retrieve the most gendered occupations per gender for each language model. More specifically, we extract the raw probabilities of pronouns with all genders per sentence first, and then calculate the average probabilities of three genders for each occupation. We get a ranking in descending order for each gender based on probabilities of that gender, and the occupations with the highest probability remain and are displayed in the plots. Finally, we plot stacked bar graphs where bars of different genders stack on each other, which helps to visualize the model's preference for pronouns of different genders.

As explained in Section 3.2, we use the single-entity version and its alternative to perform this pronoun prediction task. We only demonstrate and analyze the results of the alternative templates here, as they make sure comparison conditions are equal for occupations and participants. The results of the original single-entity version are moved to Appendix D. The average probabilities of each occupation are not normalized because this helps us have a fair observation across genders and occupations. We will use feminine, masculine, and neutral probability to refer to the probability of female, male, and neutral pronouns extracted from the model based on a specific occupation in an input sequence.

The results of base models, which are not trained at all for any tasks, will be reported firstly in Section 5.1, and then we will make a comparison over models with three versions in Section 5.2: *base* version (without any training), *adapter* version (combined with a language adapter trained for language modeling), *finetune* version (fully fine-tuned for language modeling). We will discuss the results of testing on English and Chinese data separately. We also propose a bias score as a method to help us measure gender bias, which will be elaborated on in the following contents.

5.1. Base Models

For base models, we obtain 15 most gendered occupations in total (five per gender) from each of the MLM models and the results are shown in Figure 5.1 and Figure 5.2.

5.1.1. English

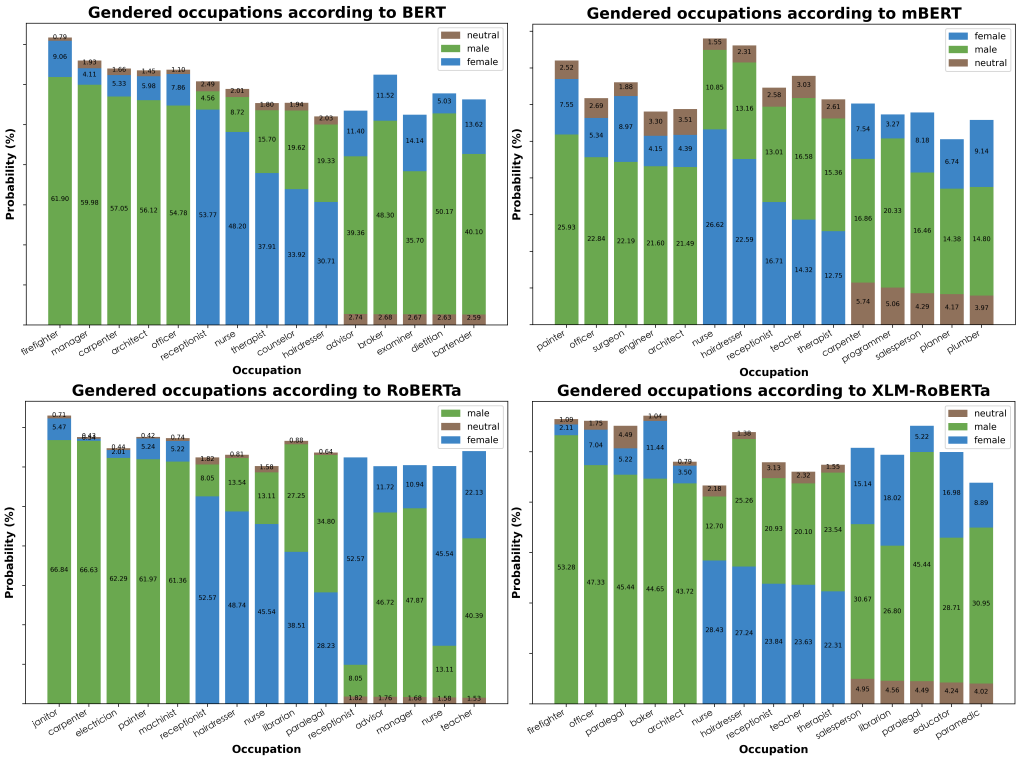


Figure 5.1.: Top 5 gendered occupations per gender across BERT, mBERT, RoBERTa and XLM-RoBERTa based on English data of the alternative single-entity version.

We will first focus on the detailed gendered occupations across the models. For masculine occupations, *officer* and *architect* are gendered occupations common to three out of four models; the frequency of *firefighter* and *painter* is two out of four; the rest are unique for each model. As for feminine occupations, *receptionist*, *nurse* and *hairdresser* are gendered occupations for all of the models; *therapist* is common to three out of four models. The mascu-

line and feminine occupations do not overlap and conflict with each other, but some of the "neutral" ones in the graphs are also masculine or feminine, and their masculine and feminine probabilities are even higher than the neutral probability. Accounting for these reasons, these neutral occupations are not considered neutral-gendered. Therefore, these language models share common feminine occupations explicitly, while the most masculine professions are model-specific. This may reflect the fact that the data used in model pretraining contain typical female stereotypes.

An interesting general observation is that the masculine probability of the most masculine occupations is very high and overwhelmingly in favor of the other two gendered probabilities, especially the monolingual models BERT and RoBERTa, all of which surpass 50%. For feminine occupations, the predominance of feminine probability decreases and is even comparable to the masculine probability in the multilingual models. Obviously, the neutral occupations we get for each model do not really tend to be neutral but tend to be masculine in most cases. We assume this might be due to *male-default* from internal gender information in language models, and this issue leads to overestimating the distribution of male instances over female or neutral ones (Stanczak and Augenstein, 2021). The male-default problem may also make many professions tend to be masculine, so that the most masculine occupations from the MLMs differ from each other.

By comparing the three types of gendered occupations in a monolingual model and its corresponding multilingual model one-to-one, we can perceive that in the bar graphs of the monolingual model, the area occupied by the bar of dominant gender is much more than that of the other two genders, whereas the gap of the bar areas of different genders in the multilingual model is relatively small. The proportion of the bar area of the neutral probability (brown) increases compared to the monolingual model. For example, for each masculine occupation in mBERT, the bar area gap among different gender probabilities is smaller than that of the same part in BERT. In addition, the sum of the probabilities of pronouns in all genders from monolingual models is higher than that of multilingual models overall, with the former being at least 50% and the latter generally less than 50%, especially for mBERT. Based on these findings, we think the degree of pronominal gender bias in the monolingual models is greater than that of the multilingual models.

To transform abstract perceptions into concrete quantitative analyses and establish a unified standard for comparing bias in both monolingual and multilingual models (accounting for variations in raw probabilities across models), we apply a ratio that helps measure the degree of gender bias, which is calcu-

lated as follows:

$$\text{ratio} = \frac{P(\text{gender_X})}{P(\text{gender_Y})} \tag{5.1}$$

There are three genders in our case, so we define the gender with the highest probability as the primary gender, the one with the lowest probability as the least gender, and the middle one as the secondary gender. We adopt the sum of the ratio between primary and secondary genders and the ratio between primary and least genders. But to keep the bias score still a probability and to make it more intuitive, we divide the sum of the ratios by two and then subtract from one. The bias score is roughly quantified this way:

$$\text{bias_score} = 1 - \tfrac{1}{2} \times \left(\frac{P(\text{secondary})}{P(\text{primary})} + \frac{P(\text{least})}{P(\text{primary})} \right) \tag{5.2}$$

Occupations	**BERT**		**mBERT**		**RoBERTa**		**XLM-RoBERTa**	
	EN	ZH	EN	ZH	EN	ZH	EN	ZH
masculine	**0.93**	**0.87**	**0.81**	**0.79**	**0.97**	**0.84**	**0.92**	**0.91**
feminine	0.79	0.57	0.59	0.68	0.76	0.60	0.55	0.67
neutral	0.83	0.61	0.64	0.63	0.84	0.80	0.71	0.70
average	0.85	0.68	0.68	0.70	0.86	0.75	0.73	0.76

Table 5.1.: **Gender bias scores** averaged on the bias scores for each group of the gendered occupations displayed in Figure 5.1 and Figure 5.2 and **averages** based on the other three gender bias scores in the same column. The highest score of each model among three groups of gendered occupations for English (EN) and Chinese (ZH) data is in **bold**. The higher score between the monolingual and multilingual model pair for each language is underlined.

The higher the score, the larger the gap among probabilities of pronouns with different genders, which implies that the model prefers the primary gender over the other two, thus reflecting pronominal gender bias to some extent. For each gendered group of occupations per model, we calculate the bias score of each occupation in the group and then average their scores as the final bias score of this gendered group. For example, the bias scores of each masculine occupation from BERT are computed and averaged as the first listed bias score in Table 5.1. But note that the bias scores based on the "neutral" occupations in the third row in Table 5.1 do not represent how much the models prefer

the neutral pronouns because the "neutral" occupations here are likely to be masculine or feminine as we explained before.

From Table 5.1, we can see that: 1) the average bias scores based on the gendered scores of BERT and RoBERTa are much higher than those of mBERT and XLM-RoBERTa, respectively; 2) the bias scores based on the masculine occupations are the highest among three groups of gendered occupations, which suggests a definite masculine probability advantage in masculine occupations and a relatively weaker feminine probability advantage in feminine occupations. Therefore, our assumptions above from the visualization have been confirmed. The multilingual MLMs seem to contain more internal pronominal gender bias than the monolingual English MLMs, and these MLMs are inherently male-oriented.

5.1.2. Chinese

From Figure 5.2, the most masculine occupations among the Chinese MLMs are 酒保 *(bartender)* and 机械师 *(machinist)* and the most feminine ones are 护士 *(nurse)* and 秘书 *(secretary)*. Similar to the results of the English MLMs, the neutral occupations are not really neutral and the neutral probability in each occupation in the graphs is nearly zero. This is because there are no official neutral pronouns in the Chinese language, the pronoun we choose for experiments is the relatively common one, but it may still not be covered by the model vocabulary.

The probability distributions from the Chinese MLMs are also masculine dominated and even more than the English MLMs. The masculine probability is always higher than the feminine probability even in some of the feminine occupations like 药剂师 *(pharmacist)* in BERT and mBERT. Therefore, the feminine occupations listed in the graphs are not really feminine like the neutral ones, but still masculine, which shows that the Chinese MLMs prefer pronouns with male gender much more over female and neutral pronouns.

Like the English MLMs, the sum of pronoun probabilities in the Chinese multilingual models (less than 25% on average) is much smaller than in the monolingual models (more than 50% on average). We assume this is because multilingual models often have larger vocabularies covering multiple languages, and this can result in a broader and more diverse set of tokens, making it less likely for the model to predict pronouns. However, different from the English models, the bias scores of the monolingual models are only slightly lower than the scores of their multilingual versions according to Table 5.1. This means that there is no big difference between the pronominal gender bias contained in the Chinese monolingual and multilingual models.

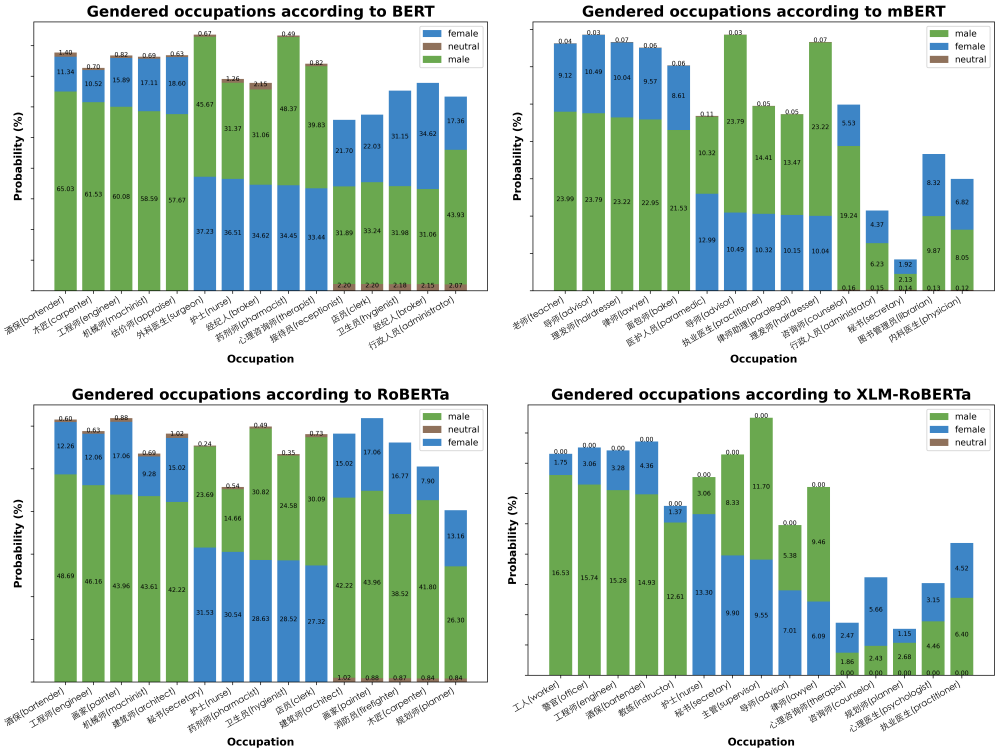


Figure 5.2.: Top 5 gendered occupations per gender across BERT-wwm, mBERT, RoBERTa-wwm and XLM-RoBERTa based on Chinese data of the alternative single-entity version.

5.2. Comparison across the three model versions

We extract the three most gendered occupations for each gender from each of the three versions of the multilingual models mBERT and XLM-RoBERTa. The gendered occupations and their probabilities are displayed in Figure 5.3 and Figure 5.4.

5.2.1. English

As we can see from Figure 5.3, for the multilingual models mBERT and XLM-RoBERTa, the most intuitive feeling is that no matter which group of masculine/feminine/neutral occupations, the gendered occupations obtained from the base model versions result in higher summed pronoun probabilities than those of the adapter and finetune versions. We speculate that this may be re-

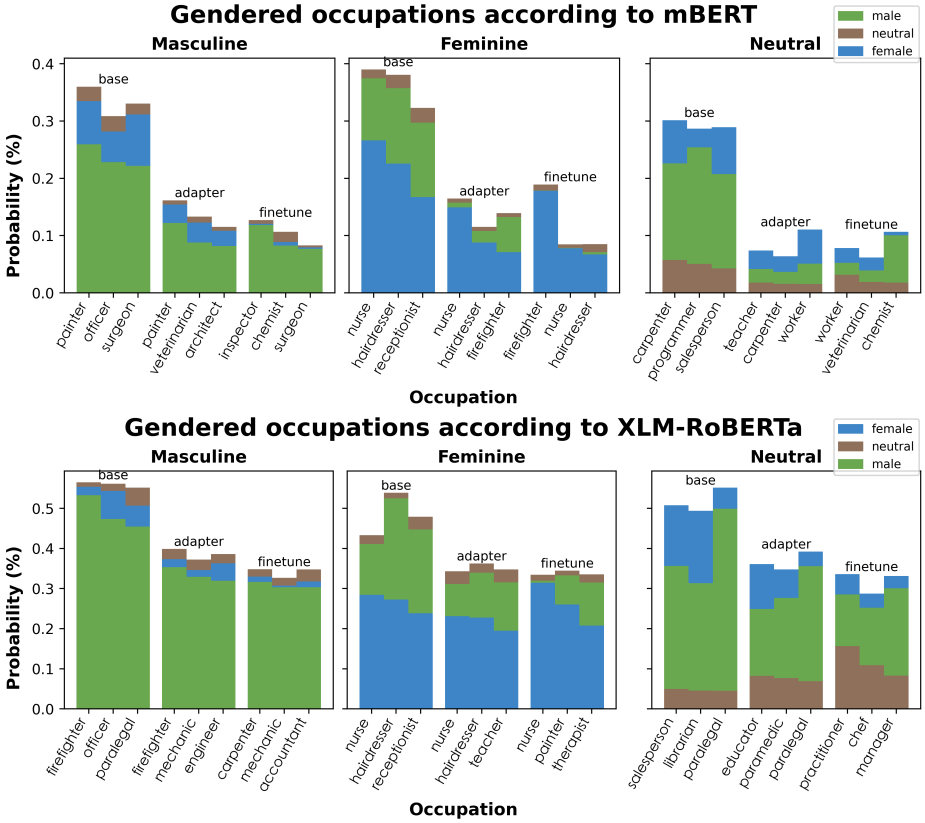


Figure 5.3.: Top 3 gendered occupations per gender across the three versions of mBERT and XLM-RoBERTa testing on English data of the alternative single-entity version.

lated to our training data used for language modeling. Although the multilingual models were also pre-trained with masked language modeling, the training data is only available in English in our case, which may contain different linguistic patterns and its size is smaller than the pre-training data. Therefore, our trained models might not generalize as well to pronouns in some contexts and might adapt to different linguistic nuances.

Like the base version, the neutral occupations from the adapter and finetune versions of mBERT and XLM-RoBERTa are not really neutral-gendered but even masculine in general, because their masculine probabilities are larger than the feminine and neutral ones as plotted in Figure 5.3. Even though the probability distributions of the gendered occupations identified in the adapter and finetune versions look close, the specific occupations do not entirely align, pos-

sibly due to different sizes of the training parameters. However, there are still overlapped gendered occupations across the three model versions. In mBERT, relatively common masculine occupations are *painter* and *surgeon*, while in the XLM-RoBERTa, they are *firefighter* and *mechanic*. These occupations appear in the top three for two out of the three model versions. In contrast, the most feminine occupations exhibit more consistency across all three model versions of both mBERT and XLM-RoBERTa. Occupations such as *nurse* and *hairdresser* consistently rank in the top three across all model variants. Building on this observation, we can get the same conclusion as in Section 5.1.1: these multilingual language models share similar feminine occupations explicitly, while the most masculine professions are model-specific. This trend may stem from the training data containing prominent and consistent female stereotypes. Conversely, for masculine occupations, as the language models generally favor males, there may be more diversity in the options, leading to variations in masculine occupations across different model versions.

In addition, we also calculate gendered and average bias scores based on the gendered occupations from the adapter and finetune versions of mBERT and XLM-RoBERTa. The computed bias scores are showcased in Table 5.2. We can find that the bias scores based on masculine occupations from every model version trained on English data are generally higher than those based on the feminine occupations. This tells us that the masculine probability advantage in masculine occupations is stronger than the feminine probability advantage in female occupations, and the adapter version and the finetune version are still consistently male-oriented as the base version. It is also important to note that the bias scores obtained in Table 5.2 based on the neutral occupations do not represent the extent to which the model favors the neutral pronouns, since the masculine or feminine probabilities are even higher than the neutral probabilities in their probability distributions.

5.2.2. Chinese

Compared to the results obtained from training on English data, the outcomes from training on Chinese data exhibit similarities and language-specific nuances. Analyzing Figure 5.4, we observe that, firstly, in contrast to XLM-RoBERTa trained on English data, the sum of pronoun probabilities derived from gendered occupations in its adapter and finetune versions trained on Chinese data is larger than that of the base version. But for mBERT, gendered occupations from the base version still result in higher summed pronoun probabilities.

Secondly, for mBERT and XLM-RoBERTa trained on Chinese data, mascu-

Occupations	mBERT_EN			XLM-RoBERTa_EN			mBERT_ZH			XLM-RoBERTa_ZH		
	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune
masculine	**0.81**	**0.79**	**0.93**	**0.92**	**0.92**	**0.95**	**0.79**	**0.89**	**0.82**	**0.91**	**0.89**	**0.94**
feminine	0.59	0.77	**0.93**	0.55	0.69	0.83	0.68	0.68	0.72	0.67	0.86	0.75
neutral	0.64	0.42	0.42	0.71	0.62	0.55	0.63	0.82	0.80	0.70	0.81	0.83
average	0.68	0.66	0.76	0.73	0.74	0.78	0.70	0.80	0.78	0.76	0.85	0.84

Table 5.2.: **Gender bias scores** averaged on the bias scores for each group of the Top 5 gendered occupations and **averages** based on the other three gender bias scores in the same column. The left section displays results from the models trained on English (EN) data, while the right section displays results from the models trained on Chinese (ZH) data. The highest score of each model version among three groups of gendered occupations is in **bold**. The highest average score among the three versions per model for each language is underlined.

line occupations from different model versions show little overlap, while feminine occupations are relatively consistent. For instance, 护士 *(nurse)* consistently ranks as the most feminine occupation across all three versions of XLM-RoBERTa, aligning with findings from Section 5.1.1. However, it is noted that some occupations listed as "feminine" in Figure 5.4 are masculine in fact, especially the "feminine" occupations extracted from mBERT. To be more specific, only 医护人员 *(paramedic)* in mBERT-base is feminine for real, while other 'feminine" occupations such as 律师 *(lawyer)* from mBERT-adapter and mBERT-finetune, result in higher predicted masculine probabilities. This suggests that the Chinese training data in our case may exhibit more bias towards the masculine pronouns and make the trained models more male-dominated.

Finally, the neutral probability based on all gendered occupations shown in Figure 5.4 is almost zero because of the absence of officially recognized neutral pronouns in the Chinese language as explained in Section 5.1.2. The neutral occupations would certainly not be considered truly neutral-gendered either.

As indicated in Table 5.2, for each model version trained on Chinese data, the bias scores from the masculine occupations are also higher compared to those derived from the feminine occupations. Additionally, regardless of whether the models are trained on English or Chinese data, the highest average bias score among the three versions of mBERT or XLM-RoBERTa is always not from the base version. Therefore, we assume that additional training could introduce more gender bias, which probably originates from our specific training data.

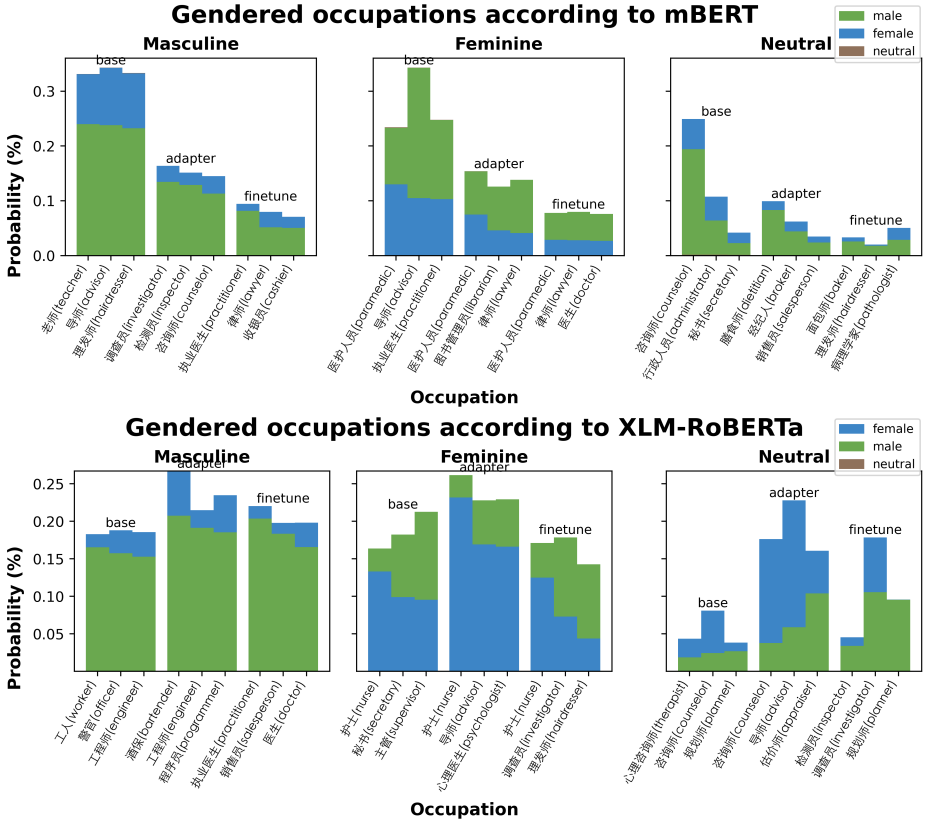


Figure 5.4.: Top 3 gendered occupations per gender across the three versions of mBERT and XLM-RoBERTa testing on Chinese data of the alternative single-entity version.

Chapter 6.

Coreference Resolution

As introduced in Chapter 3, our approach for the coreference resolution task involves utilizing MLM and generative models and employing exact metrics to specifically measure the ability of the models to perform this task. In this chapter, we will initially present and analyze the accuracy and F1 scores achieved by the MLM and generative models without additional training. We will then compare and discuss the different results arising from different version variants of the Winogender Schema, such as the single-entity and the no pronoun versions. After that, we will make a comparison of the gender bias and the results obtained from the three versions of the multilingual models (mBERT and XLM-RoBERTa). These versions include the *base* version (without any training), *adapter* version (incorporating a language adapter trained for language modeling), and *finetune* version (fully fine-tuned for language modeling). The findings will be presented separately for English and Chinese, as some of the conclusions may be language-specific.

6.1. Base Models

In the case of base models, we utilize the original pre-trained models without doing any additional training. We will assess these models using different versions of data and different types of prompts and then compare the results.

6.1.1. English

Table 6.1 demonstrates the average accuracy and F1 scores of MLM and generative language models based on various prompt template types, *Yes or No*, *Direct*, *Option* and *Who*. But before obtaining the final results shown in Table 6.1, we also do some preliminary experiments to explore better-formulated prompts. We take into account two factors that might have impacts on the performance: 1) the definite article "the" at the end of a sentence could serve as a clue for language models (we slightly modify the English Winogender dataset

Prompt	MLM				Generative
	Verbalized		Non-verbalized		Non-verbalized
	accuracy	F1	accuracy	F1	accuracy
Yes or No	50.24	39.68	45.78	37.36	16.73
Direct	52.17	43.35	25.02	22.51	27.22
Option	49.77	40.28	41.73	34.04	30.69
Who	51.32	42.94	34.90	28.37	44.26
Who (no-pronoun)	51.20	42.71	38.33	31.69	52.64

Table 6.1.: Accuracy and F1 scores of MLM (BERT, mBERT, RoBERTa and XLM-RoBERTa) and generative (T5, mT5 and FLAN-T5) models tested on English data of the double-entity and no-pronoun version prompted with the templates of *Yes or No*, *Direct*, *Option* and *Who*. The listed results of each prompt template type are the average based on all of the used language models.

by replacing "someone" with "the person" when test sentences end with "the"); 2) quotes are also likely to influence the outputs. To detect the influence of these two factors, we try prompts with four permutations of including or excluding "the" or quotes. It has been found that they lead to varying degrees of impact in different directions for different models. For instance, excluding quotes in prompts generally decreases the T5's performance for all prompt template types but increases the mT5's performance for the template types *Direct* and *Option*. It is hard to unify excluding or including "the" or quotes for all prompts, so we preserve all variants but only report the one with the best performance per prompt template type in Table 6.1.

For the results of the MLM models in Table 6.1, all accuracy of verbalized predictions is around 50%, which suggests that the MLM language models might just randomly guess from two options. The performance is even much worse when taking non-verbalized predictions. The generative models also perform poorly on this task and the highest accuracy is just around 50%. In addition, the performance of *Who (no-pronoun)*, where the pronoun in each sentence is replaced with the entity itself that this pronoun refers to, is much better than that of *Who* for the generative model, but only a few units better for the MLM models. Therefore, the models are still unable to predict perfectly correctly even if the referent entity is already known. Concerning these findings, we conclude that both the MLM and generative models are quite bad at coreference resolution in our case and they do not really understand prompted test sentences from Winogender Schema when doing a zero-shot transfer with-

out any training, even though these sentences are unambiguous and easy to resolve.

We also observe that there is a lot of variation in performance across different prompt templates, and this can be a future work where further experiments on exploring the correlations between prompt length and performance can be conducted. However, it needs to be careful that some values used for evaluation may be meaningless. We should pay attention to detailed outputs by models and check if these outputs make sense, as metrics are computed based on them. The accuracy and F1 scores of non-verbalized predictions from the MLM models based on *Yes No*, for example, are the highest among these prompt template types. But this does not imply that the MLM models actually understand the meaning of "yes" or "no" in a specific test sentence, because some models like BERT indiscriminately generate "no" based on each different test sample.

Model	Single-Entity		Double-Entity			
	Non-Verbalized		Verbalized		Non-Verbalized	
	accuracy	F1	accuracy	F1	accuracy	F1
BERT	15.08	16.77	51.10	40.88	19.39	16.36
mBERT	19.02	20.77	50.08	40.36	16.17	14.40
RoBERTa	32.26	36.38	52.87	45.07	26.53	23.59
XLM-RoBERTa	10.43	10.70	50.90	37.98	15.46	13.36
T5	29.05				13.83	
mT5	11.41				6.62	
FLAN-T5	51.59				52.18	

Table 6.2.: Accuracy and F1 scores of the MLM (BERT, mBERT, RoBERTa and XLM-RoBERTa) and generative (T5, mT5 and FLAN-T5) models tested on English data of the original single-entity and double-entity version prompted with the templates of *Yes or No*, *Direct* and *Who*. The listed results of each model are the average based on all of the used prompted templates.

Table 6.2 shows the results of the MLM and generative models tested on the single-entity and double-entity data. Since *Option* prompt templates require at least two entities as options, they cannot be used to prompt the single-entity data. To make a fair comparison between the performance tested on the single-entity and the double-entity data, the results in Table 6.2 are based on the test data only prompted with the templates of *Yes or No*, *Direct* and *Who*, without *Option*.

We expect the model performance to be better tested on the single-entity data than on the double-entity data, as we assume the models could be confused by two different entities in a sentence more easily than only one single entity. However, the results of BERT, XLM-RoBERTa and FLAN-T5 seem different from what we expected. We figure out that the performance gap between single-entity and double-entity of these three models all originate from prompt templates of *Yes or No*. For the single-entity test data, correct answers of *Yes or No* prompts are all "yes" because there is only one entity in each sample and only this entity can be referred to. While for the double-entity data, the gold labels could be "yes" or "no" because both of the entities have the chance to be referred to. As mentioned above, some models cannot understand the exact meaning "yes" or "no" and predict "no" for every different input sentence. Therefore, the *all-"no"* predictions are all wrong if they are evaluated against the gold labels of the single-entity data, but could happen to be right if evaluated against the answers of the double-entity data. For the results on the single-entity test data prompted with the other two prompt types *Direct* and *Who*, the MLM and generative models indeed perform better as we expected.

Generally speaking, the multilingual language models perform worse overall compared with the corresponding monolingual models according to Table 6.2. FLAN-T5 is an exception since it has already been fine-tuned on over 1,000 more tasks including definite pronoun resolution which is similar to our coreference resolution task. We also review the raw outputs from the generative models. The generations from FLAN-T5 are of the best quality because they are short and there are only direct and specific nouns, not necessarily the occupation or the participant in the input sentence, but no other redundant or meaningless texts. The quality of generations from mT5 is much better than T5 even though the accuracy of T5 is much higher. Many generations by T5 simply repeat a few consecutive words in the original tested sentence, and they might happen to be right in the evaluation. Besides, during the preliminary phase, we experiment with several lengths of output texts and find that the accuracy of T5 drops a lot if the length is increased from 10 to 15. This further confirms that the outputs of T5 repeat more words including both the occupation and participant and the quality is quite bad. We will discuss evaluation standards for the performance of the generative models in more detail in Chapter 7.

6.1.2. Chinese

We make preliminary experiments with variations on Chinese prompts with regard to Chinese linguistic features. One is whether or not to include quota-

Prompt	MLM				Generative
	Verbalized		Non-verbalized		Non-verbalized
	accuracy	F1	accuracy	F1	accuracy
Yes or No	50.00	33.33	31.64	22.67	34.32
Direct	65.87	60.82	16.67	17.90	8.99
Option	73.68	70.11	12.41	13.91	19.34
Who	56.81	49.07	0.45	0.68	10.21
Who (no-pronoun)	59.22	51.33	0.63	0.78	27.19

Table 6.3.: Accuracy and F1 scores of MLM (BERT-wwm, mBERT, RoBERTa-wwm and XLM-RoBERTa) and generative (ChatLM and mT5) models tested on Chinese data of the double-entity and no-pronoun version prompted with the templates of *Yes or No*, *Direct*, *Option* and *Who*. The listed results of each prompt template type are the average based on all of the used language models.

tion marks in a prompt. The other is to add "the answer is" at the end of input sentences to hint models further. Preliminary experimental results show that the hint words "the answer is" help the models to understand test data. Still, quotation marks have different effects on different types of language models and prompts. We keep the results for each type of prompt variant, but Table 6.3 only shows the best results for each prompt under this prompt type.

Similar to the English data test results, for non-verbalized predictions, the performance of the MLM models based on the test sentences prompted by *Yes or No* type is the best among the prompt types listed in Table 6.3, but this seems superficial. For the MLM models, the accuracy based on verbalized outputs is 50% and the F1 score is around 30.33%. It is clear that the models do not understand the meaning of "是 (yes)" and "否 (no)". Because all verbalized predictions are "是 (yes)", and the probability of generating "是 (yes)" is at least 100x higher than that of generating "否 (no)". We also find that "否 (no)" never appear in non-verbalized predictions, which are either "是 (yes)" or some other irrelevant words.

Unlike the results of the MLM models tested on the English data, the accuracy and F1 scores of the prompt types *Direct* and *Option* based on the verbalized predictions are well above 50%. This indicates that the models tend to make correct choices, although they may not precisely generate exact match results because the results based on non-verbalized predictions are not good enough. For *Who* and *Who (no-pronoun)* prompts, metrics based on non-verbalized predictions are nearly zero, much lower than results for other

prompt types. In contrast to English models, Chinese MLM models struggle with answering a question from the beginning but prefer completing a fragmented sentence.

The generative models appear to perform better than the MLM models, especially for *Who* and *Who (no-pronoun)*. Despite the large improvement in *Who (no-pronoun)* compared to *Who*, the model accuracy is still suboptimal. We speculate that the poor performance may be due to two factors, one is the lack of prompts with high quality, and the other is our automatic evaluation criteria.

For generative models, the accuracy of *Yes or No* seems to be the highest. However, after confirming the generated texts, we find that generative models, especially ChatLM, tend to generate "是 (yes)" like the MLM models. Additionally, "yes" and "is" share the same character "是" in Chinese and "is" is a frequently included word in generations. This issue could impact the automatic evaluation results a lot. For outputs based on the inputs prompted by *Yes or No*, we make efforts in the automatic evaluation to distinguish between "yes" and "is" based on their usage characteristics in Chinese. A comprehensive analysis of the automatic evaluation criteria will be presented in Chapter 7.

In summary, the MLM and generative models without any training struggle to comprehend Chinese Winogender sentences well based on the results in Table 6.3. To maintain parallelism with the English test data, we only adopted these specific prompt template types. However, the performance of both types of models needs improvement. Future exploration could focus on identifying prompts that better matched the models, enabling a more accurate assessment of a model's understanding of test sentences. This is crucial since low-quality prompts can introduce noise and affect the models' performance on this task.

Table 6.4 presents the comprehensive performance of each model tested on the single-entity and double-entity data prompted by different prompt types. As mentioned earlier, the prompt type *Option* is not included because it does not apply to the single-entity data. Comparing the results based on non-verbalized predictions between the two test data versions, models generally perform better on the single-entity data as expected, showing that two entities can interfere with the model's understanding.

Overall, the performance of the monolingual models is generally superior to the multilingual models. XLM-RoBERTa exhibits a large performance gap compared to other models, as its predictions often consist of meaningless symbols rather than specific text. This might be because one input sentence into other models is inserted with multiple mask tokens, and the final output is a concatenation of the individual characters corresponding to these masks. In

Model	Single-Entity		Double-Entity			
	Non-Verbalized		Verbalized		Non-Verbalized	
	accuracy	F1	accuracy	F1	accuracy	F1
BERT-wwm	41.41	45.51	62.34	53.28	24.99	23.71
mBERT	32.18	35.79	54.92	44.00	16.77	13.29
RoBERTa-wwm	63.23	65.22	62.20	52.67	34.46	28.32
XLM-RoBERTa	4.17	6.44	50.32	37.19	1.75	2.12
ChatLM	52.68				22.96	
mT5	22.36				14.55	

Table 6.4.: Accuracy and F1 scores of the MLM (BERT-wwm, mBERT, RoBERTa-wwm and XLM-RoBERTa) and generative (ChatLM and mT5) models tested on Chinese data of the original single-entity and double-entity version prompted with the templates of *Yes or No*, *Direct* and *Who*. The listed results of each model are the average based on all of the used prompted templates.

contrast, each input into XLM-RoBERTa contains only one single mask token, making it more likely to output meaningless marks. Considering these factors, the performance of XLM-RoBERTa is much lower than other models. Nonetheless, the primary factor contributing to this gap is the limited capabilities of XLM-RoBERTa for coreference resolution and it may not have been pre-trained for this task, as XLM-RoBERTa also performs worse than other models when tested on English data.

A meaningful Chinese word may consist of several characters, and we increase the output sequence length from 10 to 15 for generative models to generate sufficient text for a proper evaluation of model performance. However, the automatic evaluation of generative models remains a persistent challenge in our research. High accuracy does not necessarily imply good generation quality or understanding of input sentences by the models. Our current automatic evaluation mainly revolves around judging whether the generation includes the gold label. But many generations simply repeat the input sequence, and generations may easily include that entity in the case of single-entity data with only one entity. This leads to being correct without necessarily providing a valid answer.

Taking ChatLM as an example for specific analysis, it is a small Chinese chat model. If we increase the sequence length to 100, it can generate a long text analyzing how to solve coreference resolution. This analysis may include both the occupation and the participant from the input sentence, but our existing

evaluation methods cannot achieve a semantic understanding of a long text. Therefore, extracting only a small portion of the potential generation as the final prediction, and applying a physical inclusion verification strategy without considering its semantic meaning, cannot accurately assess the performance of generative models in this task. This issue needs to be resolved in the future.

6.2. Comparison across the three model versions

In this section, our main focus will be on the comparative discussion of gender bias across the three versions of mBERT and XLM-RoBERTa, supplemented by a comparison of task performance.

6.2.1. English

Gender	mBERT						XLM-RoBERTa					
	Accuracy			F1 scores			Accuracy			F1 scores		
	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune
Female	19.66	16.82	13.75	20.94	18.64	16.68	10.81	5.96	5.57	10.83	6.59	5.96
Male	19.87	17.24	16.77	21.50	19.02	21.11	10.60	5.86	5.36	10.69	6.49	5.81
Neutral	17.53	17.11	12.86	19.14	19.14	15.68	9.87	6.07	5.21	10.15	6.75	5.78
Average	19.02	17.06	14.46	20.52	18.93	17.83	10.43	5.96	5.38	10.56	6.61	5.85
Gap	0.995	**0.156**	1.539	0.926	**0.195**	2.191	0.370	**0.069**	0.127	0.269	0.091	**0.073**

((a)) Single-entity

Gender	mBERT						XLM-RoBERTa					
	Accuracy			F1 scores			Accuracy			F1 scores		
	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune
Female	16.45	11.78	16.42	14.36	11.01	16.52	16.03	6.72	9.88	13.66	6.10	9.05
Male	16.64	11.87	17.53	14.65	11.19	17.62	15.31	6.08	9.17	13.17	5.64	8.38
Neutral	15.42	11.20	15.27	13.72	10.70	15.31	15.04	6.77	8.22	12.98	6.25	8.16
Average	16.17	11.62	16.41	14.24	10.97	16.48	15.46	9.09	6.52	13.27	8.53	5.99
Gap	0.501	**0.281**	0.755	0.348	**0.177**	0.780	0.379	**0.295**	0.582	0.259	**0.234**	0.346

((b)) Double-entity

Table 6.5.: Accuracy and F1 scores based on non-verbalized predictions of the three versions of mBERT and XLM-RoBERTa tested on English data of the original single-entity and double-entity version prompted with the templates of *Yes or No*, *Direct* and *Who*. The listed results per gender of each model are the average based on all of the used prompted templates. The smallest gap for each model among the three versions is in **bold**.

Table 6.5 presents the performance of mBERT and XLM-RoBERTa based on inputs containing different gender pronouns. We evaluate the model's performance on each gender, and the average across genders represents the overall model performance since our test data is evenly distributed among genders. A comprehensive performance gap is derived based on individual gender performances and their averages, serving as a measure of gender bias.

The performance gap for mBERT is generally higher than that of XLM-RoBERTa, but this does not necessarily imply that mBERT is inherently more biased than XLM-RoBERTa. If a model struggles with the coreference resolution task due to its limited capabilities, it may generate incorrect answers regardless of the gender of pronouns in the sentences. This results in similar performances across the three genders, obscuring any potential gender bias. Hence, a poor task performance may not accurately reflect gender bias, and our analysis primarily revolves around the results from mBERT.

From Table 6.5, we observe that the finetune version of mBERT (mBERT-finetune) generally exhibits a larger performance gap compared to the base and adapter versions (mBERT-base and mBERT-adapter). This suggests that finetuning introduces more gender bias into the model. Upon closer inspection of the model's performance across genders, it is evident that the model performs best on sentences with male pronouns, indicating a preference for males. mBERT-finetune achieves average F1 scores of 17.83 and 16.48 on the single-entity and double-entity data, respectively. The decrease in performance is mainly attributed to a bigger drop in the F1 score for male compared to the other two genders, from 21.11 (single-entity) to 17.62 (double-entity). Building on the conclusion from the previous subsection, sentences with two entities have a negative impact on the pre-trained model's performance in this task, and this also holds for both adapter and finetune versions. It indicates that these two model versions are confused by two entities, especially evident in the test samples including male pronouns. Consequently, when mBERT has the ability to understand the sentence, it identifies more with the case where the pronoun is male than the other two genders and makes the correct judgment accordingly.

Taking the example sentence from Figure 6.1, sourced from the single-entity version prompted by *Yes or No* type, generated by mBERT-finetune. As mentioned in Section 6.1, when using *Yes or No* prompts with English data, the models consistently choose "no" between the two options. However, here, mBERT-finetune outputs the correct answer "yes" only when the pronoun in the input sentence is male ("he"), indicating that the model recognizes the pronoun "he" as referring to the occupation "mechanic." This example further confirms mBERT's preference for male pronouns.

'The mechanic called to say that she had completed the repair.' In the previous sentence, does the pronoun 'she' refer to the 'mechanic'? Yes or No? no

'The mechanic called to say that he had completed the repair.' In the previous sentence, does the pronoun 'he' refer to the 'mechanic'? Yes or No? yes

'The mechanic called to say that they had completed the repair.' In the previous sentence, does the pronoun 'they' refer to the 'mechanic'? Yes or No? no

Figure 6.1.: An example from the outputs of mBERT-finetune based on test sentences from the English single-entity version data prompted by *Yes or No*. The outputs are highlighted.

From Table 6.5, we also observe that mBERT-adapter achieves the smallest performance gap across both test data versions. The results from the single-entity version show that mBERT-adapter has better performance on average than mBERT-finetune, but its performance gap is much smaller than that of mBERT-finetune. This is mainly due to its performance on the neutral gender narrowing the gap with the other two genders. Therefore, we consider mBERT-adapter to exhibit the least gender bias in English coreference resolution tasks compared to the other two versions.

However, contrary to our expectations, for both mBERT and XLM-RoBERTa, the adapter and finetune versions perform overall worse than the base versions. In other words, despite the models receiving additional training for English language modeling, their performance on coreference resolution tasks is not as good as the base version without any additional training. We will provide a detailed explanation and analysis of this result in Chapter 7.

6.2.2. Chinese

Table 6.6 presents the overall performance of different versions of mBERT and XLM-RoBERTa on the Chinese single-entity and double-entity data, as well as the specific performance of each gender and the gaps between them. Note that we list two gaps here: the inclusive gap, which integrates the results of all three genders, and the binary gap, which is solely the performance gap between male and female results.

We add the binary gap because the gender bias reflected by the inclusive gap is not accurate for several reasons. First, as emphasized earlier, there is no official neutral pronoun in Chinese. Also, based on the models' results in the pronoun prediction task, as shown in Figure 5.4, mBERT and XLM-RoBERTa generate Chinese neutral pronouns with probabilities close to zero. Therefore, there is doubt about whether mBERT and XLM-RoBERTa truly under-

Gender	mBERT						XLM-RoBERTa					
	Accuracy			F1 scores			Accuracy			F1 scores		
	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune
Female	36.10	23.71	35.80	38.25	29.87	38.53	2.23	11.10	25.57	3.05	13.48	27.45
Male	34.36	24.39	34.51	37.35	31.42	38.06	3.11	11.40	25.76	4.40	13.76	27.49
Neutral	26.10	9.811	31.29	29.19	14.42	34.25	7.16	17.61	25.30	10.48	18.49	27.39
Average	32.18	19.31	33.86	34.93	25.23	36.94	4.17	13.37	25.54	5.98	15.24	27.45
Gap (inclusive)	4.057	6.330	1.717	3.827	7.210	1.797	1.995	2.828	0.160	2.999	2.164	0.034
Gap (binary)	1.742	**0.682**	1.288	0.901	1.547	**0.474**	0.871	0.303	**0.189**	1.355	0.277	**0.044**

((a)) Single-entity

Gender	mBERT						XLM-RoBERTa					
	Accuracy			F1 scores			Accuracy			F1 scores		
	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune	base	adapter	finetune
Female	18.69	14.38	18.20	13.83	12.63	14.06	1.33	4.13	13.28	1.29	4.02	10.00
Male	18.14	15.40	19.45	13.90	13.68	15.24	1.72	4.45	13.45	1.71	4.10	9.94
Neutral	13.48	7.16	15.87	10.85	8.15	13.99	2.20	7.90	13.26	2.76	6.15	10.31
Average	16.77	12.31	17.84	12.86	11.49	14.43	1.75	5.49	13.33	1.92	4.75	10.08
Gap (inclusive)	2.193	3.434	1.313	1.343	2.226	0.540	0.299	1.604	0.080	0.561	0.928	0.152
Gap (binary)	**0.549**	1.024	1.250	**0.073**	1.051	1.178	0.396	0.321	**0.171**	0.415	0.074	**0.061**

((b)) Double-entity

Table 6.6.: Accuracy and F1 scores based on non-verbalized predictions of the three versions of mBERT and XLM-RoBERTa tested on Chinese data of the original single-entity and double-entity version prompted with the templates of *Yes or No*, *Direct* and *Who*. The listed results per gender of each model are the average based on all of the used prompted templates. The smallest binary gap for each model among the three versions is in **bold**.

stand neutral pronouns in the coreference resolution task. Secondly, whether tested on the single-entity or double-entity version, we can see that the results of the base and the adapter versions from mBERT and XLM-RoBERTa on neutral gender are contradictory. For mBERT, the performance with neutral pronouns is much lower than with the other two genders, while it is higher for XLM-RoBERTa. So we think the results based on the neutral pronouns in Table 6.6 are not that reliable, making the inclusive gap unreliable as well. Concerning this, we introduce the binary gap between only male and female results. However, this approach itself is biased, but we have to adopt it to better analyze the differences in bias among the three versions. We also look forward to the emergence of widely accepted neutral pronouns using Chinese characters instead of the Roman alphabet "TA". This would provide data including neutral pronouns to train multilingual models, enabling them to have a more comprehensive understanding of gender.

Similar to the analysis of gender bias based on English data, our focus for the bias analysis based on Chinese data also remains primarily on mBERT. This choice is due to the poor performance of XLM-RoBERTa and this may lead to unreliable performance gaps. Even though the XLM-RoBERTa-finetune results on the single-entity version seem to show great improvement compared to XLM-RoBERTa-base and XLM-RoBERTa-adapter, we find that this improvement comes from generating "是 (yes)" for every input prompted by *Yes or No*. Therefore, good results do not necessarily mean that the model can truly perform the coreference resolution task. Hence, we do not compare and analyze gender bias among the three versions of XLM-RoBERTa here.

The performance gap exhibited by three versions of mBERT across different test data versions does not show a consistent ranking pattern. Surprisingly, the results from mBERT-base and mBERT-finetune based on the single-entity version with female pronouns are better than those with male pronouns. This contradicts the results of these two model versions tested on the double-entity version and all results of the three versions of XLM-RoBERTa. Thus, we cannot rank and compare the performance gaps exhibited by the three mBERT versions tested on the single-entity data. For testing with the double-entity data, the finetune version yields the largest performance gap, followed by the adapter version, and the base version has the smallest. We guess that the bias comes from the Chinese Wikipedia data used for language modeling training. In the Chinese context, "他" (he) is often used as the default pronoun for an unknown gender, and models could be trained to recognize this specific linguistic nuance through the training data, thus resulting in a bigger gender bias.

Next, we will focus on the model performance. According to the results in Table 6.6 based on the single-entity test data, we observe that mBERT-finetune performs better than mBERT-base, but mBERT-adapter is much worse than mBERT-base. Previous research results have indicated that adapter tuning and full fine-tuning can achieve comparative performance in language understanding and generation tasks (Houlsby et al., 2019; Lin et al., 2020). However, in our study, the performance of adapter tuning is not as expected. This suggests that the models require more of their parameters to be updated in order to effectively learn from language modeling and apply that knowledge to Chinese coreference resolution.

XLM-RoBERTa-base and XLM-RoBERTa-adapter perform better on neutral pronouns compared to pronouns of the other two genders. We figure out that this is because, XLM-RoBERTa-base and XLM-RoBERTa-adapter can output "是 (yes)" for inputs containing a neutral pronoun prompted by *Yes or No*, whereas for inputs containing female or male pronouns, they often output meaningless symbols. XLM-RoBERTa-fintune shows a large improvement in

overall performance compared to the *base* and the *adapter* versions, and it mainly narrows the gap between neutral results and the other two genders. The reduction in the gap is attributed to *Yes or No* prompts. In other words, XLM-RoBERTa-finetune no longer outputs punctuation marks for inputs containing female or male pronouns but outputs "是 (yes)", thereby improving the results for these two genders to be on par with the neutral ones. However, as discussed in 6.1, the better results for *Yes or No* do not necessarily mean that the model truly understand coreference resolution. Therefore, Chinese language modeling training does not appear to assist XLM-RoBERTa in the coreference resolution task. Similarly, for mBERT, the impact of Chinese language modeling training on its coreference resolution capability is not evident either.

Chapter 7.

Discussion

This chapter will conduct a more detailed analysis of the factors that may influence the reported results of the coreference resolution task in Section 6. We mainly categorize the influencing factors into three types: 1) training data and task; 2) prompt quality; and 3) automatic evaluation methods for model performance and measurement of gender bias.

7.1. Training data and task

Based on the results presented in Section 6, where adapter and finetune versions of the models exhibit performance lower than base versions for coreference resolution on English data, we make further exploration.

To eliminate the possibility of overfitting, we confirm the evaluation loss during adapter tuning and full fine-tuning, observing a consistent decrease throughout the training process. We save the fine-tuned model or the trained adapter at the point with the lowest evaluation loss. However, our additional training makes the models more adaptable to language modeling, but the testing task of coreference resolution is more specific. We hypothesize that the mismatch between the trained and tested tasks is one reason why adapter and finetune versions do not outperform the base version in coreference resolution.

Additionally, the impact of our training data cannot be overlooked. Training models on a specific dataset, even if it is derived from Wikipedia, can lead to domain-specific adaptation. The Winogender Schema data for coreference resolution may differ a lot from the Wikipedia data used for language modeling training. Moreover, in the pronoun prediction task, the probability sum of pronouns of all genders in adapter and finetune versions has decreased compared to the base version. We assume that this might be related to our training data which may lack sufficient linguistic patterns related to pronouns. Thus, domain mismatch could also be one of the factors.

7.2. Prompted testing data

In Chapter 6, we compare the impact of different prompt types on the performance of MLM and generative models. Overall, these prompts are more effective on English data, with generally poorer results on Chinese data. Here, we primarily discuss the results based on non-verbalized predictions.

Our summary of using different types of prompts is two-fold. Firstly, whether with English or Chinese data, using *Yes or No* prompts does not yield effective results. This is due to pre-trained MLM models generating "no" for all prompted English test sentences and "yes" for all prompted Chinese sentences. Also, it is challenging to accurately automatically evaluate generative models based on *Yes or No* prompts. Secondly, there is a big disparity in the results of *Who* prompts between English and Chinese data, and the effectiveness varies with different types of language models. For example, English generative models perform very well on the coreference resolution task for test sentences prompted by *Who*. FLAN-T5 not only achieves an accuracy of 70% but also generates text with high quality (it means that generations contain only one of the two entities excluding irrelevant words, and are not merely a repetition of the input sentence). However, when using *Who* prompts with Chinese data, it has almost no effect, and the accuracy of MLM models in resolving reference between two entities in the sentence is close to zero.

'The visitor asked the receptionist if she/he/they needed to sign in.' In the previous sentence, who needed to sign in? The receptionist

'The visitor asked the receptionist if she/he/they could estimate how long the wait would be.' In the previous sentence, who could estimate how long the wait would be? The receptionist

Figure 7.1.: An English example from the outputs of BERT based on a test sentence from the double-entity version data prompted by *Who*. The outputs are highlighted.

In addition, we calculate the percentage of outputs containing occupations or participants. For English data, regardless of the prompt type, pre-trained MLM models tend to output participants much more frequently than occupations for all test sentences. We attribute this to the positions of occupations and participants in the sentences. In most cases, occupations appear at the beginning of the sentence, which is further away from pronouns than participants. Therefore, MLM models tend to select the entity closer to the pronoun in a sentence. In the example in Figure 7.1, the participant *visitor* appears at the beginning and the occupation *receptionist* follows. However, regardless

of the gender of the pronouns or the sentence's meaning, BERT consistently outputs *receptionist* simply because it is closer to the pronouns. This finding aligns with the explanations provided by large language models, as mentioned in Kotek et al. (2023), regarding the reference resolution for pronouns. One of the main explanations is that the models consider pronouns more likely to relate to the most recently mentioned noun or the noun closest to the pronoun.

On the other hand, T5 and mT5 show a higher percentage of containing occupations compared to containing participants. As mentioned earlier in Section 6.1, these models, especially T5, tend to simply repeat the input sentence. Occupations are more likely to appear in generations as they often appear at the beginning of the input sentences. In our test data, occupations and participants each constitute 50% of the gold labels. However, the imbalanced ratio of them in the outputs indicates that the models might rely on the syntactic structure of the input sentences rather than the semantic content to solve this task for all genders.

Hence, based on the analysis above, we acknowledge that both the choice of prompts and test data have a big impact on the final assessment of pronominal gender bias.

7.3. Evaluation

The automatic evaluation method has a direct impact on the results of coreference resolution, consequently influencing our measurement of gender bias using the performance gap. As mentioned earlier in Section 6.1, automatic evaluation poses a huge challenge in our research. Our evaluation standards are generally a bit strict.

For MLM models, we primarily employ the exact match method, where the prediction must match the label perfectly to be considered correct. However, we have also encountered cases involving synonyms. For instance, in the English example in Figure 7.2, *seller* and *salesperson* are synonyms, while in the Chinese example in Figure 7.2, 机械工 *(mechanic)* and 机修工 *(mechanic)* are synonymous. Even though the model outputs semantically correct answers, they are regarded as incorrect due to the strict criteria of exact match. Only through manual evaluation can synonyms be understood and accurately judged as correct.

For generative models, the challenge mainly lies in dealing with Chinese data prompted by *Yes or No*. As mentioned earlier, in Chinese, "yes" and "is" are represented by the same character "是". Generations often include both "yes" and "is". The previous judgment criteria are ineffective in this case. De-

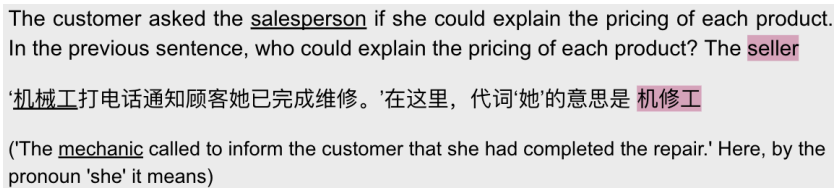


Figure 7.2.: English and Chinese examples from the outputs of BERT based on a test sentence from the double-entity version data prompted by *Who* and *Direct*, respectively. The outputs are highlighted.

termining how to differentiate between “yes” and “is” in automatic evaluation brings us trouble. We have redefined the criteria based on the main features of the text generated by mT5 and ChatLM. Despite these efforts, there may still be some correct cases overlooked, and only manual evaluation can effectively distinguish the different meanings of “是” (yes/is).

We make some attempts to relax evaluation criteria, such as the additional criteria for evaluating “former” and “latter” introduced in Section 3.5. Nonetheless, automatic evaluation still cannot cover all the specific cases in our study and only plays a role in high-quality outputs. This is because it cannot comprehend semantics and can only make general evaluations based on string matching or inclusion. Consequently, this makes it harder to accurately evaluate the model performance and analyze the gender bias exhibited by different versions of models in the coreference resolution task.

To summarize, from the selection of the training task and data to the choice of testing tasks and data, as well as how to design appropriate prompts for test data and accurately evaluate the results, each step can influence the final assessment of the performance and gender bias of different model versions.

Chapter 8.

Limitation & Future Work

The analysis in Chapter 7 reveals several limitations in our study that need improvement, both in model training and testing.

Firstly, regarding training, it is crucial to address the mismatch between training and testing tasks, as well as the domain mismatch between training and testing data. If we train the models for coreference resolution, performance would become better probably, but then that is not a fair comparison with the base version. For future investigations, efforts should be made to achieve consistency between training and testing tasks. Comparing the results of base models with post-finetuning and post-adaptation results for the same task would provide a fairer comparison.

Secondly, the quality of prompt templates needs enhancement. To ensure parallelism between English and Chinese test data, we directly translate the English prompt templates into Chinese. However, the results indicate that some prompts are not effective for Chinese data in coreference resolution. Exploring prompts that are effective for both Chinese and English could lead to outputs with better quality. This can help us evaluate the model's ability to understand the task and distinguish gender differences in pronouns. Therefore, future work should dive more into exploring high-quality and effective prompts.

Lastly, the automatic evaluation in coreference resolution is also a limitation of this study and directly impacts the measurement of gender bias. Our current evaluation method is general and applicable to the majority of predictions. How to evaluate special cases like synonyms appropriately is a big challenge and future research can further improve this aspect.

Furthermore, based on our current results, FLAN-T5 exhibits remarkable performance because it has been trained for definite pronoun resolution. This inspires the exploration of few-shot training for coreference resolution on different model versions. We hope models understand sentences rather than producing the same results for all genders. Achieving a more reliable performance gap requires models to comprehend the meaning of sentences, thus contribut-

ing to a more accurate measurement of gender bias.

Future work can make further exploration of adapters, such as experimenting with different hyperparameters during adapter training. Currently, we only use language-specific adapters and they can also be stacked with task-specific adapters (Pfeiffer, Vulić, et al., 2020). Considering the inclusion of task-specific adapters for coreference resolution in the comparison, how would the performance and gender bias of each model version change? All of these can serve as interesting future research directions.

Chapter 9.

Conclusion

Our research revolves around the Winogender dataset with parallel English and Chinese data and three versions of models of *base*, *adapter* and *finetune*. to perform the downstream tasks of pronoun prediction and coreference resolution.

For pronoun prediction, our study begins by exploring the probabilities assigned by monolingual and multilingual models to pronouns of different genders. We further compare the most gendered occupations generated by mBERT and XLM-RoBERTa across different versions. Also, we measure the extent of the models' preference for pronouns associated with a particular gender. We conclude that multilingual MLMs exhibit a higher degree of internal pronominal gender bias compared with monolingual MLMs, and all three model versions demonstrate a bias toward male-oriented pronouns in this task.

For coreference resolution, we compare the performance of pre-trained monolingual and multilingual models across different data variants and prompt types. It has been found that monolingual models generally outperformed their multilingual counterparts, and the choice of prompt type has a big impact on the results. We also conduct a comprehensive analysis comparing the performance and pronominal gender bias across the three versions of mBERT and XLM-RoBERTa. We find that the *finetune* version exhibits a higher performance gap among the three genders compared to the *base* and the *adapter* versions. This suggests a bigger gender bias in the *finetune* version which prefers male pronouns. While the *adapter* version displays a smaller performance gap than the *finetune* version, but its performance is generally inferior in most cases. This reflects both the disparity in performance between adapter tuning and full fine-tuning for coreference resolution, and also the smaller absorption of pronominal gender bias by the *adapter* version from the training data.

However, our study still needs further improvement. For instance, the exploration of prompts in coreference resolution needs to be more extensive, and the current automatic evaluation of predictions is insufficient to cover all cases.

The overall model performance is still bad and this directly negatively influences the accurate measurement of gender bias. Finally, there is ample room for in-depth research into different adapter architectures and their impact on gender bias. How to improve the model performance and how to choose more appropriate training data as well as downstream tasks to compare the gender bias included in the three versions should be solved in future work. Finally, there are some other different architectures for adapters, and the research on the gender bias of adapters can be further deepened.

Appendix A.

Prompt Templates

Appendix A. Prompt Templates

Prompt Type	English Prompt Template
Yes or No	1. '[TEXT]' In the previous sentence, does the pronoun '[PRON]' refer to '[MENTION1]'? Yes or No? 2. '[TEXT]' Here, by the pronoun '[PRON]' it means '[MENTION1]'? Yes or No? 3. '[TEXT]' I think the pronoun '[PRON]' means '[MENTION1]'. Yes or No? 4. [TEXT]' Here, does the pronoun '[PRON]' stand for '[MENTION1]'? Yes or No? 5. '[TEXT]' In the previous sentence, can the pronoun '[PRON]' be replaced with '[MENTION1]'? Yes or No?
Direct	1. '[TEXT]' In the previous sentence, the pronoun '[PRON]' refers to 2. '[TEXT]' Here, by the pronoun '[PRON]' it means 3. '[TEXT]' I think the pronoun '[PRON]' means 4. '[TEXT]' Here, the pronoun '[PRON]' stands for 5. '[TEXT]' In the previous sentence, the pronoun '[PRON]' can be replaced with
Option	1. '[TEXT]' In the previous sentence, does the pronoun '[PRON]' refer to '[MENTION1]' or '[MENTION2]'? 2. '[TEXT]' What does the pronoun '[PRON]' in the above sentence refer to? '[MENTION1]' or '[MENTION2]'? 3. 'In the sentence below, does the pronoun '[PRON]' stand for '[MENTION1]' or '[MENTION2]'? '[TEXT]' 4. '[TEXT]' Replace the pronoun '[PRON]' in the above sentence with the correct option: - '[MENTION1]' - '[MENTION2]'
Who	1. '[TEXT]' In the previous sentence, who/whose [Action]? The

((a)) Prompt templates in English with double entities used in Coreference Resolution

	Chinese Prompt Template
Yes or No	1. '[TEXT]' 在前一句中，代词 '[PRON]' 是指代 '[MENTION1]' 吗？是或者否？ 2. '[TEXT]' 在这里，代词 '[PRON]' 的意思是 '[MENTION1]' 吗？是或者否？ 3. '[TEXT]' 我认为代词 '[PRON]' 的意思是 '[MENTION1]'。是或者否？ 4. '[TEXT]' 在这里，代词 '[PRON]' 是否代表 '[MENTION1]'？是或者否？ 5. '[TEXT]' 在前一句中，代词 '[PRON]' 是否可被替换为 '[MENTION1]' 吗？是或者否？
Direct	1. '[TEXT]' 在前一句中，代词 '[PRON]' 指的是 2. '[TEXT]' 在这里，代词 '[PRON]' 的意思是 3. '[TEXT]' 我认为代词 '[PRON]' 是指 4. '[TEXT]' 在这里，代词 '[PRON]' 代表的是 5. '[TEXT]' 在前一句中，代词 '[PRON]' 可以替换成
Option	1. '[TEXT]' 在前一句中，代词 '[PRON]' 是指代 '[MENTION1]' 还是 '[MENTION2]'？答案是 2. '[TEXT]' 代词 '[PRON]' 在上述句子中指代什么？'[MENTION1]' 还是 '[MENTION2]'？答案是 3. 在下面的句子中，代词 '[PRON]' 代表的是 '[MENTION1]' 还是 '[MENTION2]'？'[TEXT]' 答案是 4. '[TEXT]' 上句中的代词 '[PRON]' 可以替换为下列哪个符合原句意的词语？- '[MENTION1]' - '[MENTION2]' 答案是
Who	1. '[TEXT]' 在前一句中，谁 [Action]？

((b)) Prompt templates in Chinese with double entities used in Coreference Resolution

Table A.1.: Parallel English and Chinese prompt templates with double entities in Coreference Resolution. **[TEXT]**: the original sentence from Winogender Schema; **[PRON]**: the pronoun in the original sentence; **[MENTION1]**: the entity *occupation* in the original sentence; **[MENTION2]**: the entity *participant* in the original sentence.

Appendix B.

Entity dictionary between English and Chinese

English	Chinese	English	Chinese
technician	技术员	accountant	会计
supervisor	主管	engineer	工程师
worker	工人	educator	教育工作者
clerk	店员	counselor	咨询师
inspector	检测员	mechanic	机械工
manager	经理	therapist	心理咨询师
administrator	行政人员	salesperson	销售员
receptionist	接待员	librarian	图书管理员
advisor	导师	pharmacist	药剂师
janitor	看门人	psychologist	心理医生
physician	内科医生	carpenter	木匠
nurse	护士	investigator	调查员
bartender	酒保	specialist	专科医生
electrician	电工	officer	警官
pathologist	病理学家	teacher	老师
lawyer	律师	planner	规划师
practitioner	执业医生	plumber	水管工
instructor	教练	surgeon	外科医生
veterinarian	兽医	paramedic	医护人员
examiner	检验员	chemist	化学师
machinist	机械师	appraiser	估价师
nutritionist	营养师	architect	建筑师
hairdresser	理发师	baker	面包师
programmer	程序员	paralegal	律师助理
hygienist	卫生员	scientist	科学家
dispatcher	调度员	cashier	收银员
auditor	审计员	dietitian	膳食师
painter	画家	broker	经纪人
chef	厨师	doctor	医生
firefighter	消防员	secretary	秘书
customer	顾客	taxpayer	纳税人
employee	员工	client	客户
pedestrian	行人	student	学生
patient	病人	homeowner	房主
teenager	青少年	undergraduate	本科生
visitor	来访者	child	孩子
advisee	被指导者	onlooker	围观者
witness	证人	protester	抗议者
victim	受害者	resident	居民
owner	主人	passenger	乘客
buyer	买方	bystander	旁观者
guest	客人	someone	某人
the person	某人		

Table B.1.: The entity dictionary of occupations between English and Chinese.

Appendix C.

Hyperparameters

Parameter	Value
do_train	True
do_eval	True
validation_split_percentage	10
per_device_train_batch_size	8
per_device_eval_batch_size	8
max_seq_length	512
evaluation_strategy	steps
eval_steps	10,000
save_steps	150,000
max_steps	150,000
num_train_epochs	5
save_total_limit	2
load_best_model_at_end	True
overwrite_output_dir	True

Table C.1.: Parameters of training language adapters and full fine-tuning models

Table C.1 displays hyperparameter settings of training language adapters in our case. Parameters that are not mentioned here are set to their default values.

Appendix D.

Stacked Bar Plots of Gendered Occupations

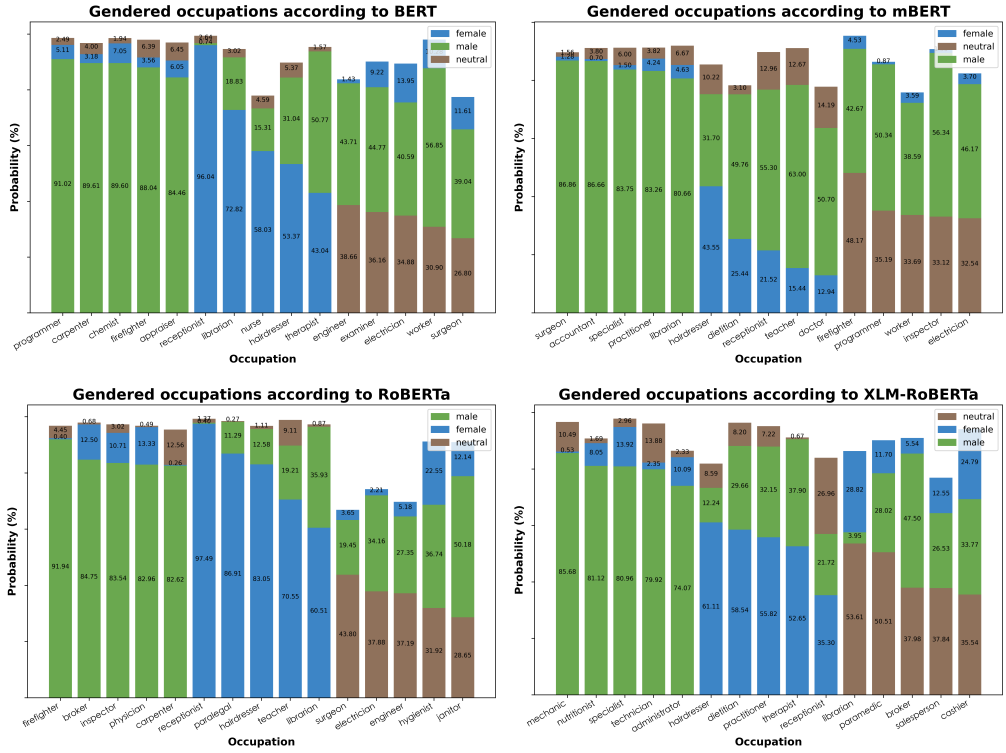


Figure D.1.: Top 5 gendered occupations per gender across BERT, mBERT, RoBERTa and XLM-RoBERTa based on English data of the original single-entity version.

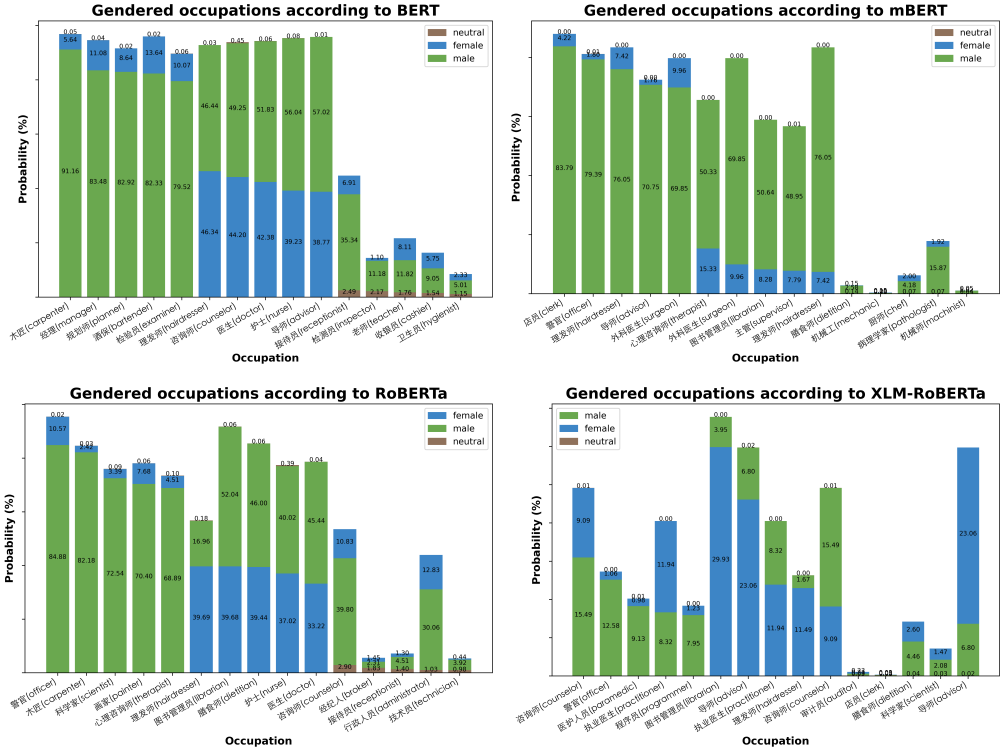


Figure D.2.: Top 5 gendered occupations per gender across BERT-wwm, mBERT, RoBERTa-wwm and XLM-RoBERTa based on Chinese data of the original single-entity version.

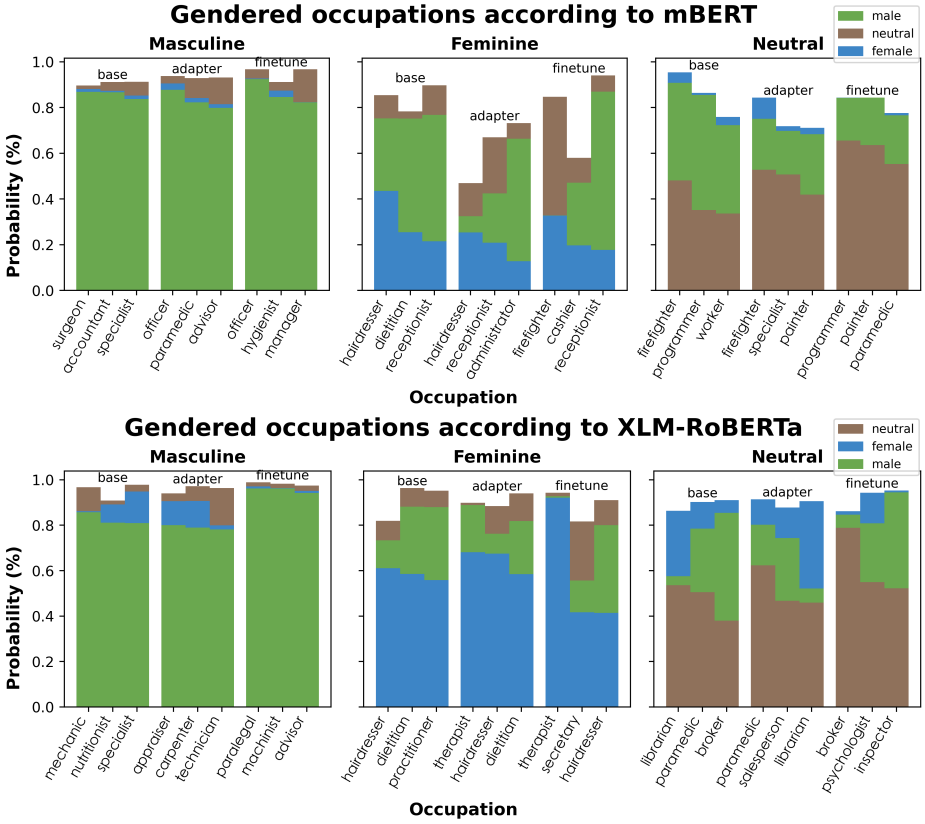


Figure D.3.: Top 3 gendered occupations per gender across the three versions of mBERT and XLM-RoBERTa testing on English data of the original single-entity version.

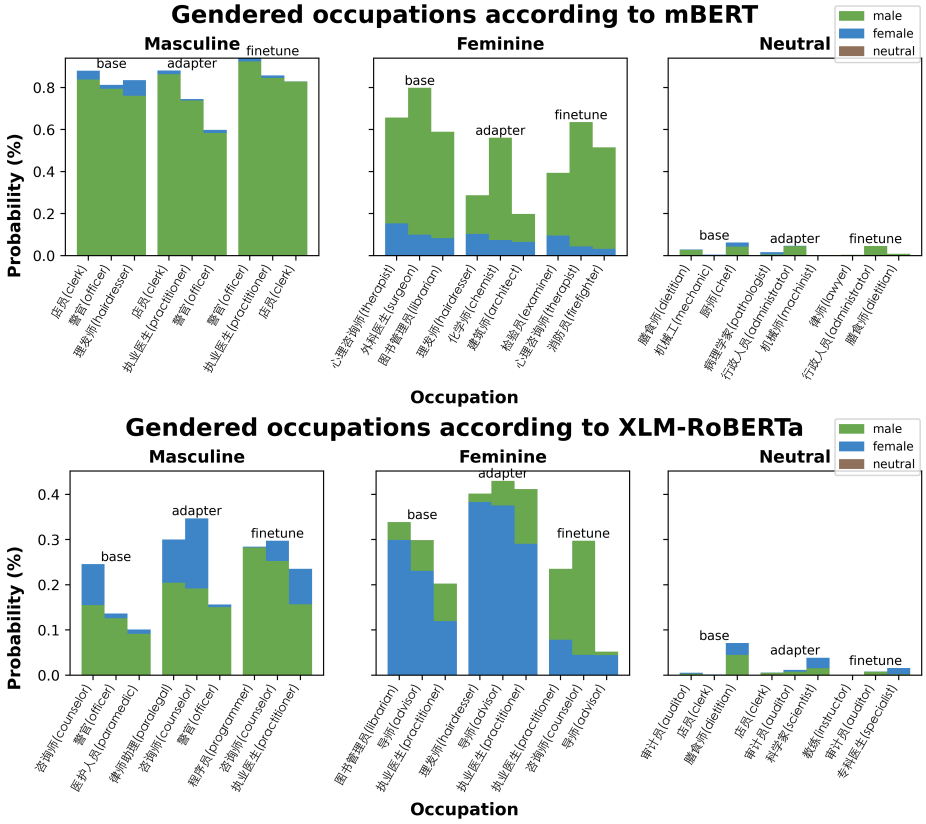


Figure D.4.: Top 3 gendered occupations per gender across the three versions of mBERT and XLM-RoBERTa testing on Chinese data of the original single-entity version.

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